

NO NO₂!

High resolution Spatial & Temporal Nowcasting of
Tropospheric NO₂ via TEMPO Satellite

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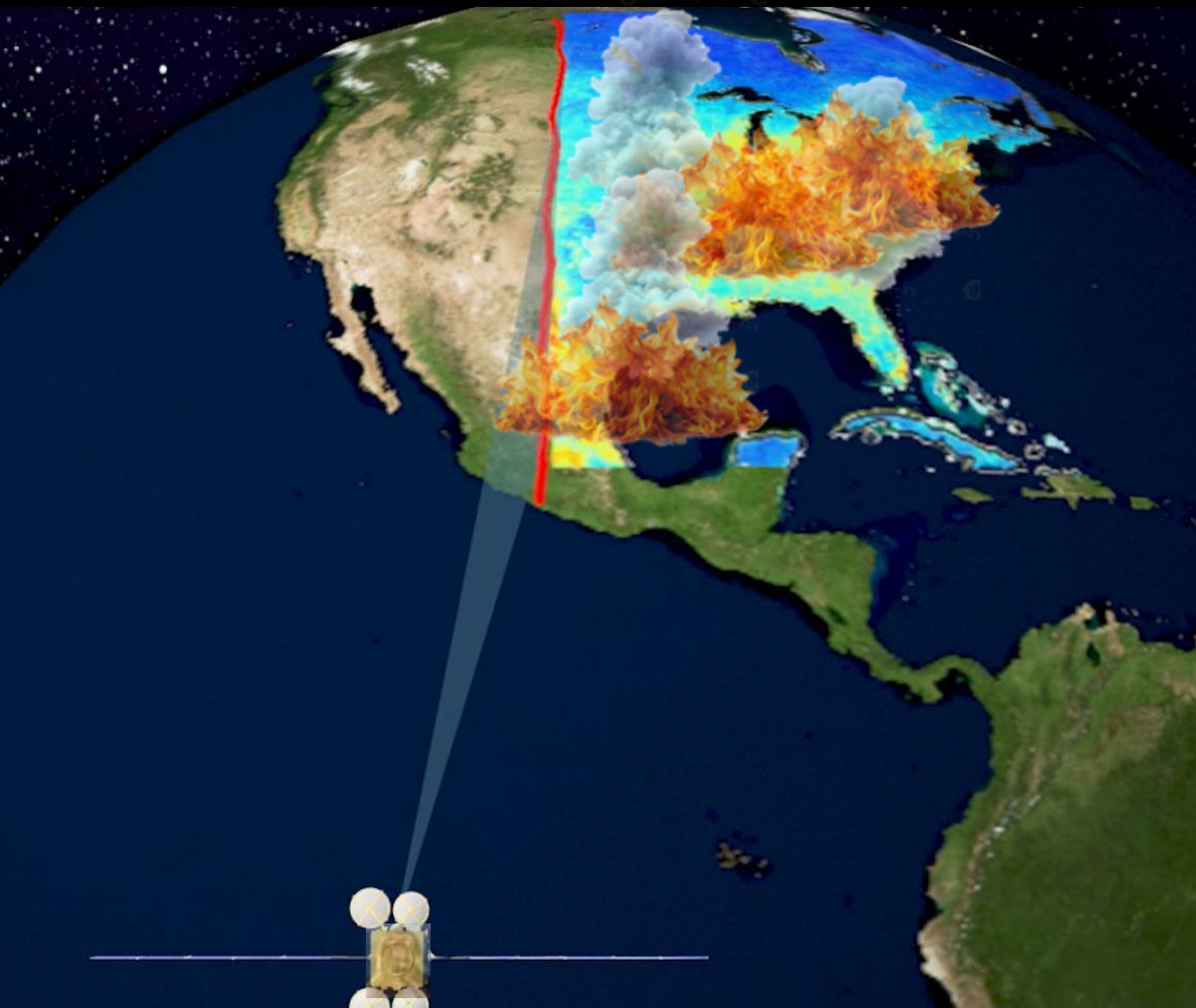
Mentor: Kyo Lee
Group Supervisor: Michael Turmon
August 21st, 2026

For required markings, please visit <https://mh.jpl.nasa.gov>

*Image credit: NASA,
Vecteezy*



Agenda



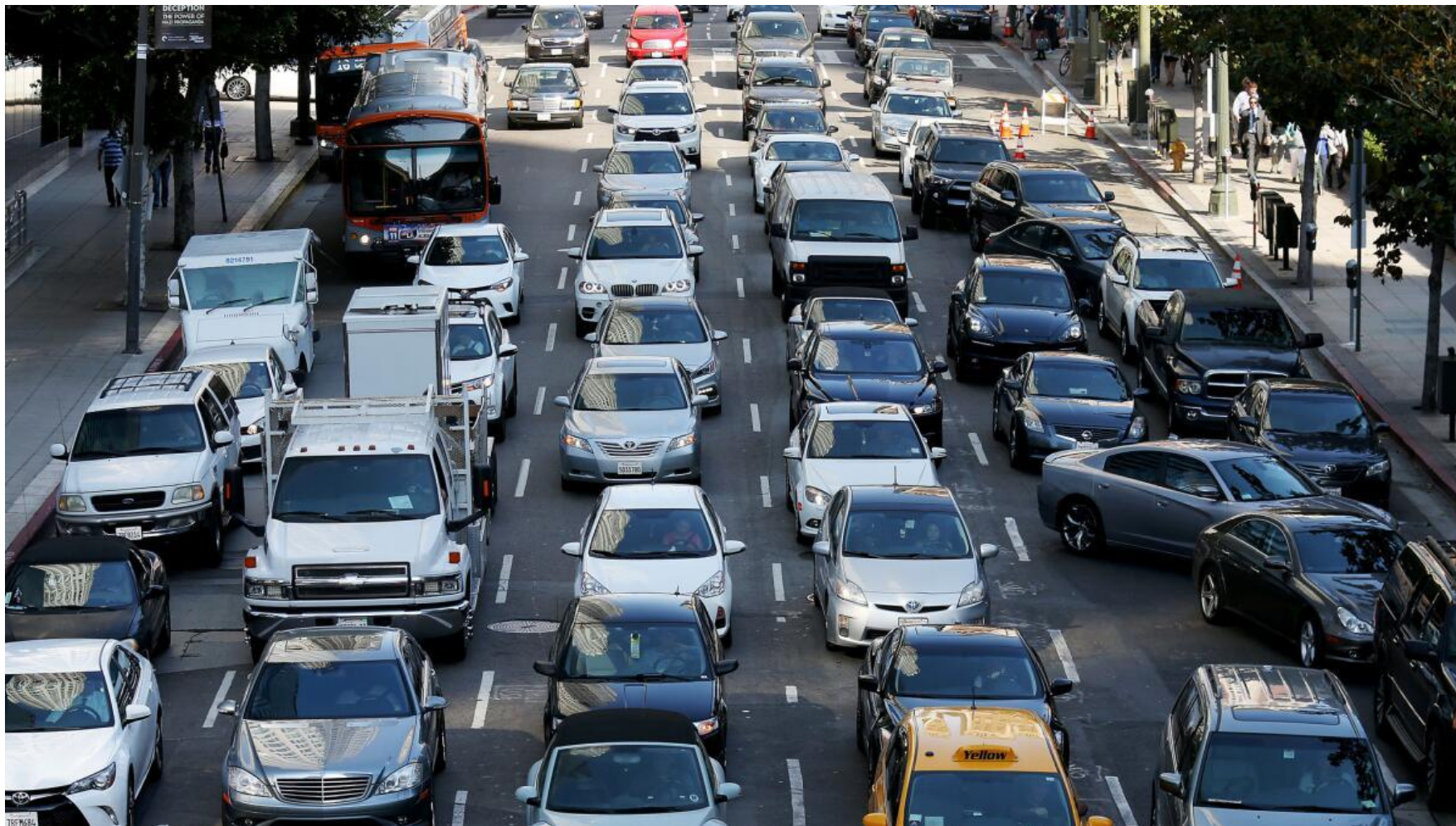
- 01** Motivation & Context
- 02** Architecture
- 03** Methodology & Design
- 04** Results
- 05** Discussion & Analysis
- 06** Conclusion & Future

Image credit: NASA EARTHDATA, Vecteezy

01 Motivation & Context



01 Motivation & Context

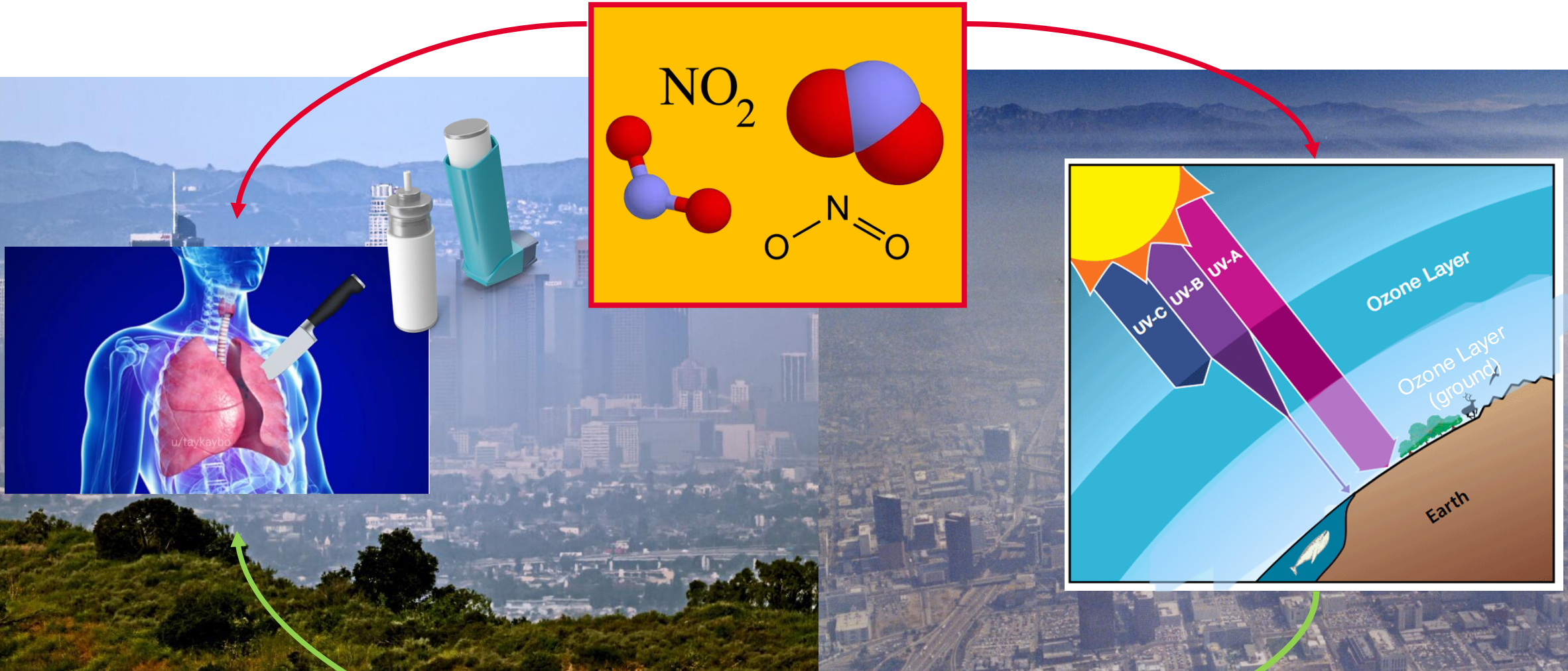




DANGER: RESPIRATORY DAMAGE



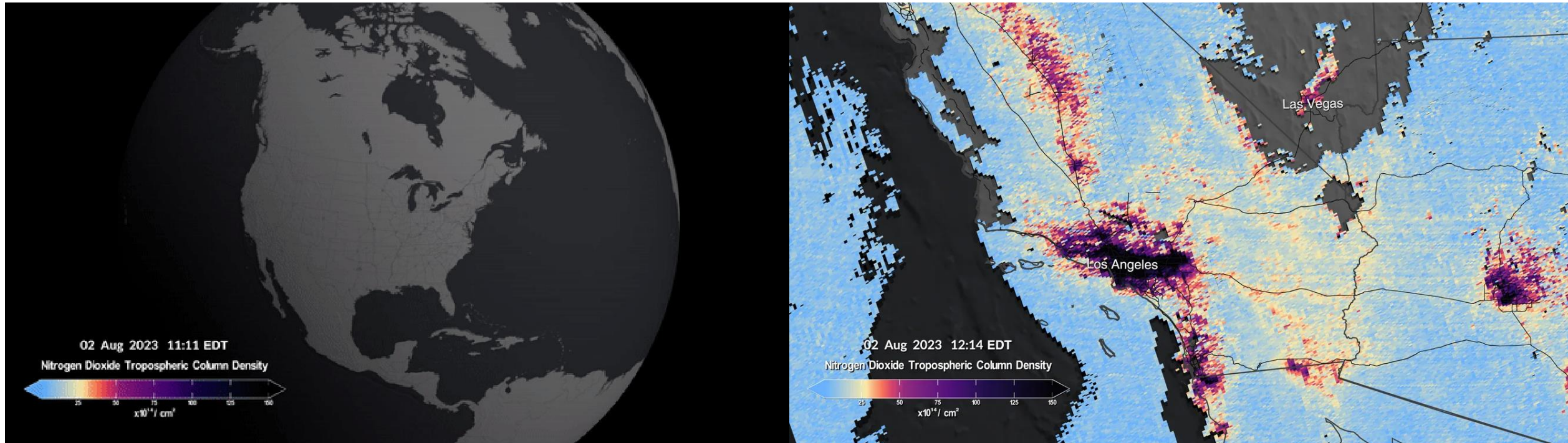
—
This becomes incredibly dangerous to people with respiratory issues!



What Are We Looking At?

Satellite:

TEMPO View → Hourly updates, total spatial coverage, but coarse resolution ($\sim 2.0 \times 4.5$ km per pixel).



What Do We Know is True?

Ground Truth:

EPA Ground Stations → High accuracy, ground-level truth, but **sparse spatial coverage** (only 66 stations across all of California).

Satellites see everywhere with low resolution

Ground Stations see high resolution, but almost nowhere

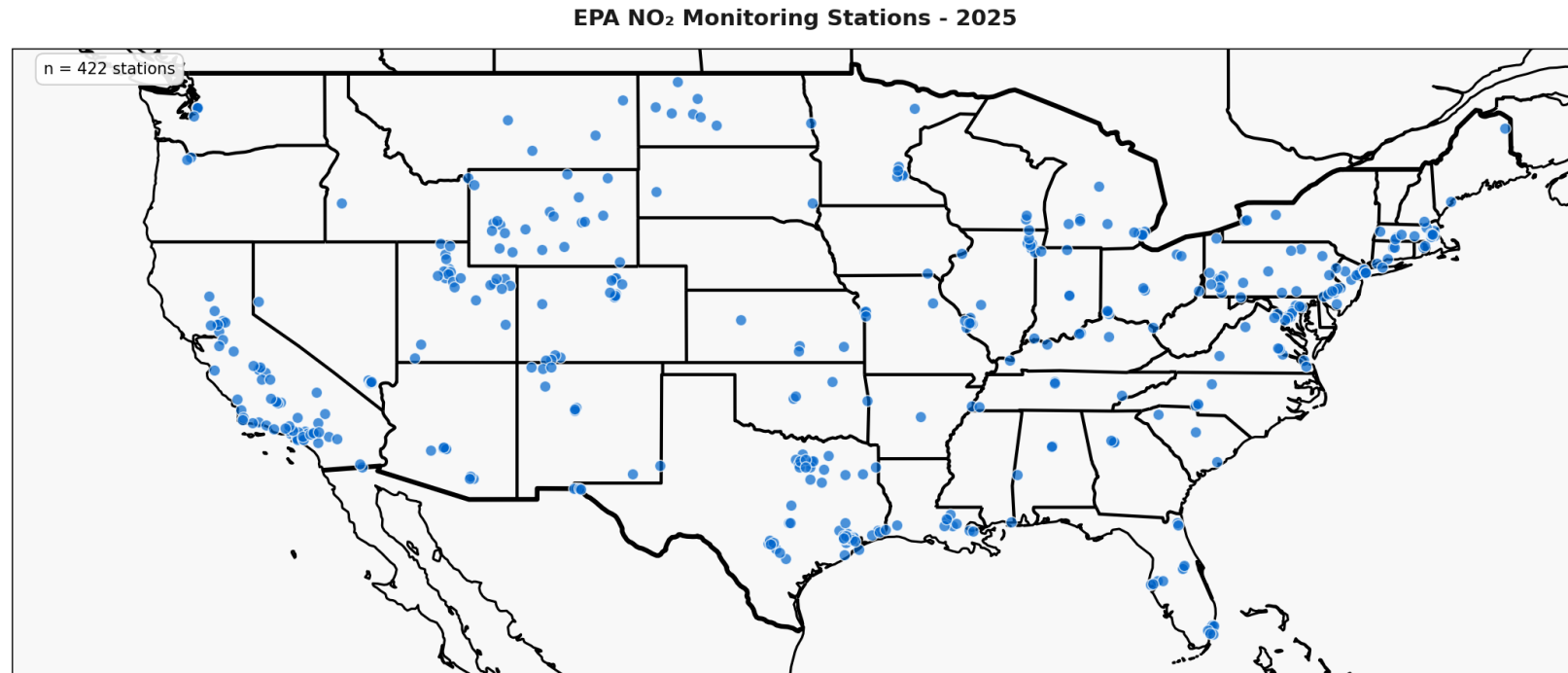
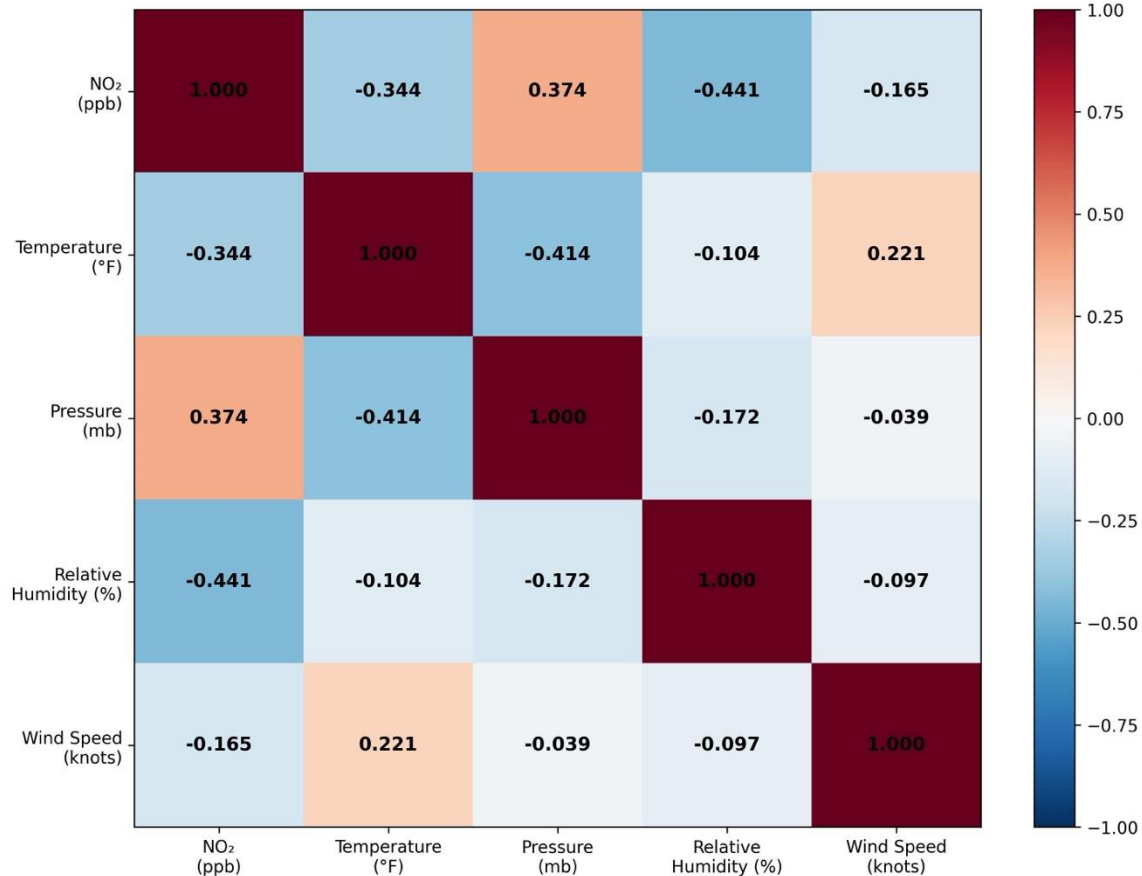


Image Credit: DNREC – Delaware.gov

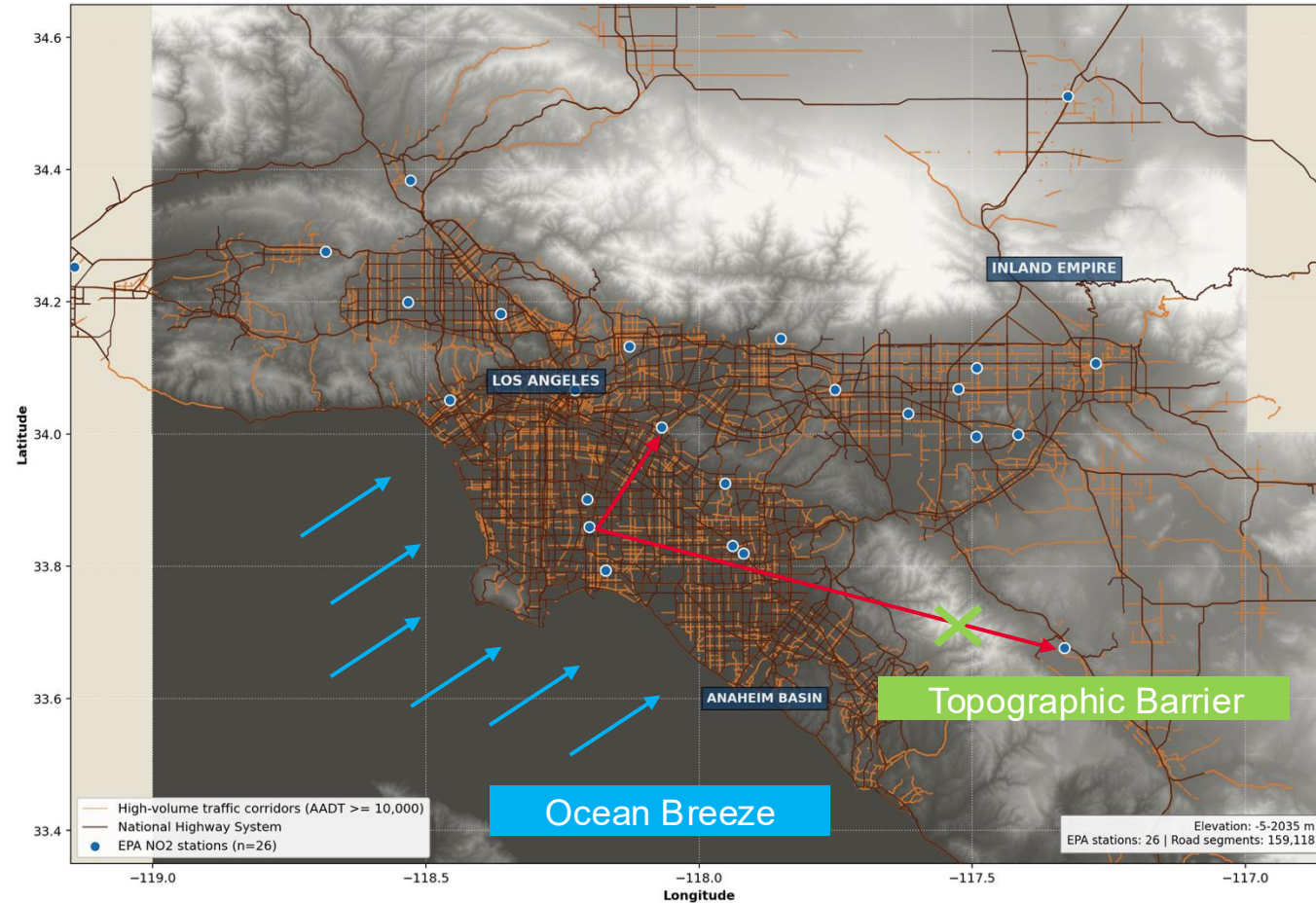
02 Architecture

The Goal: Building the Digital Twin

Correlation Matrix: NO₂ and Meteorological Variables
Compton Site 2023-2024

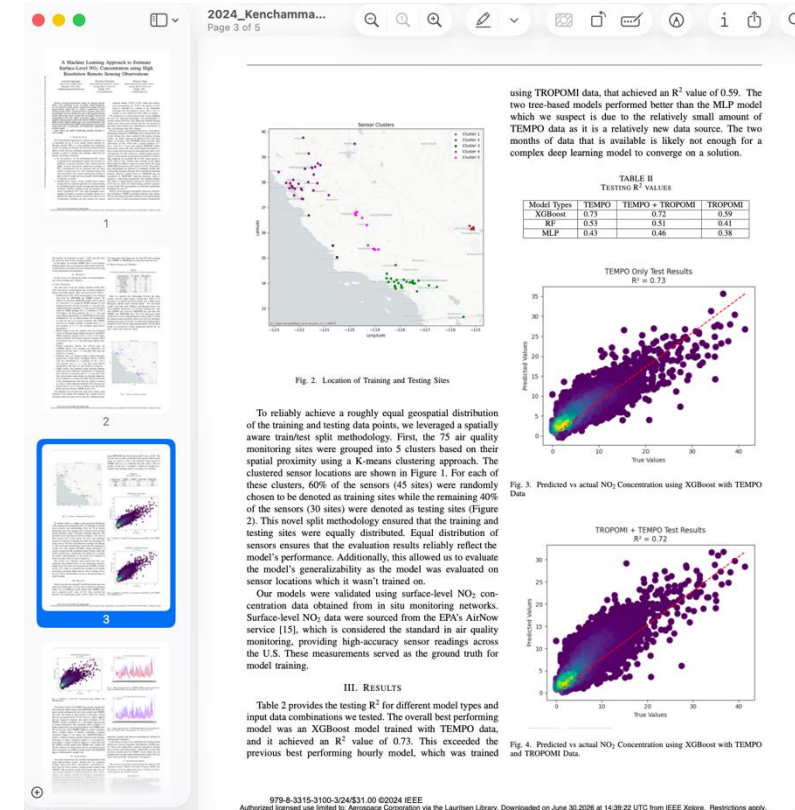


EPA NO₂ Monitoring Stations: Inland Empire, Greater Los Angeles, and Anaheim Basin
SRTM elevation with traffic-volume roads and highways | 2024 station network



Machine Learning Approach to Estimate Surface-Level NO₂ Concentration using High Resolution Remote Sensing Observations ; A. Kenchamma ; 2024

- 2024 IEEE paper on estimating AirNow NO₂ using machine learning in California.
- Previous research used ~ 2 months of early TEMPO data. We have access to ~ 14 months of Level-3 V03 TEMPO NO₂ data.
- They used XGBoost as their main model



What is XGBoost?

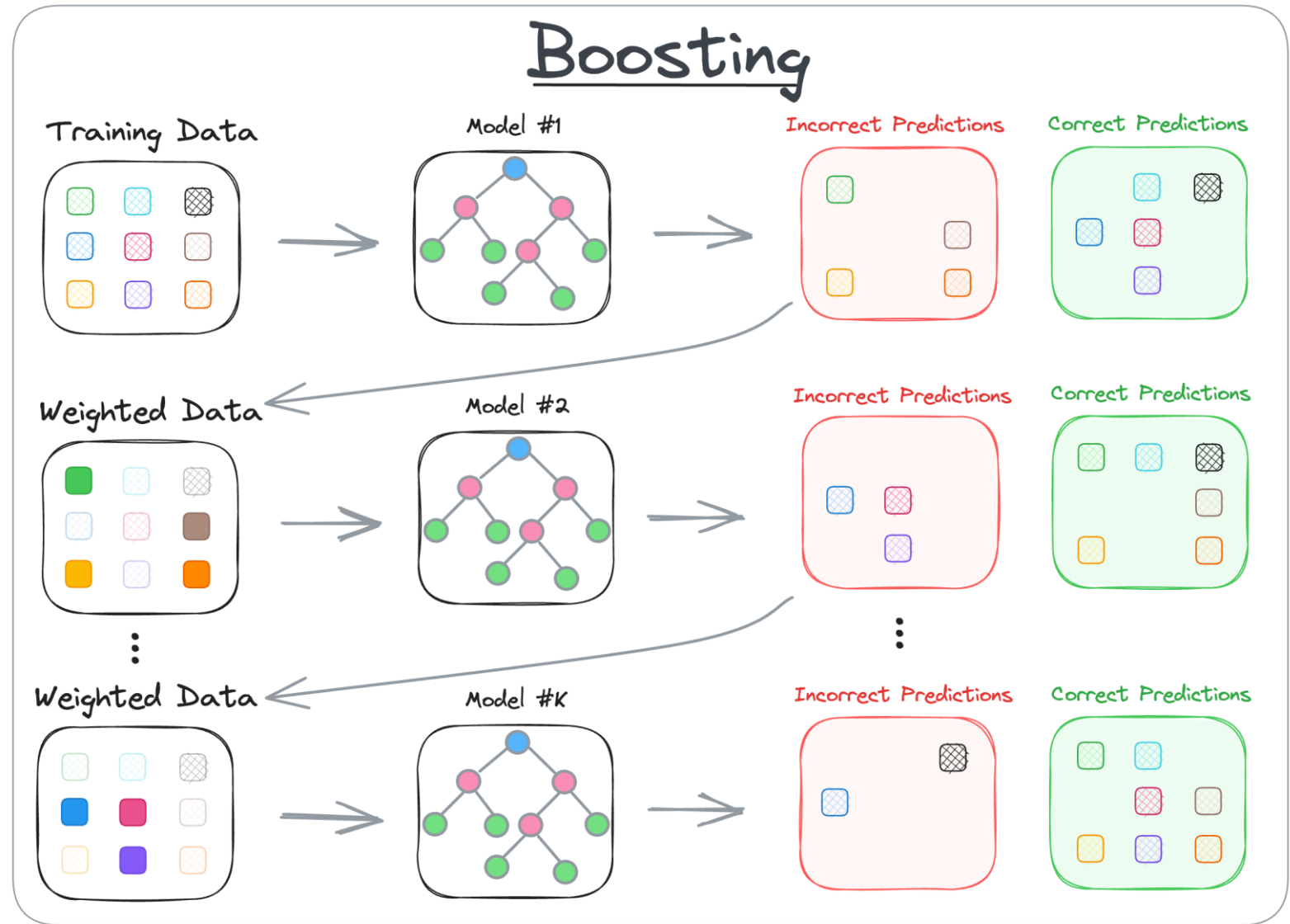
A decision tree asks simple yes/no questions:

- “Is the boundary layer < 300 m?”

Very fast at training because of the **Yes/No** logic conditions (~20 minute training)

TEMPO + Metrological + Terrain + Traffic + Time as “Training Data”

AirNow Ground Stations as “Correct Predictions”



Predictors That We Are Using

Satellite (3)

TEMPO_NO2
tempo_local_max
tempo_local_spatial_variance

Meteorology (7)

t2m
d2m
u10
v10
sp
blh
thermal_inversion_proxy

Photochemistry (2)

solar_zenith_angle
tcc

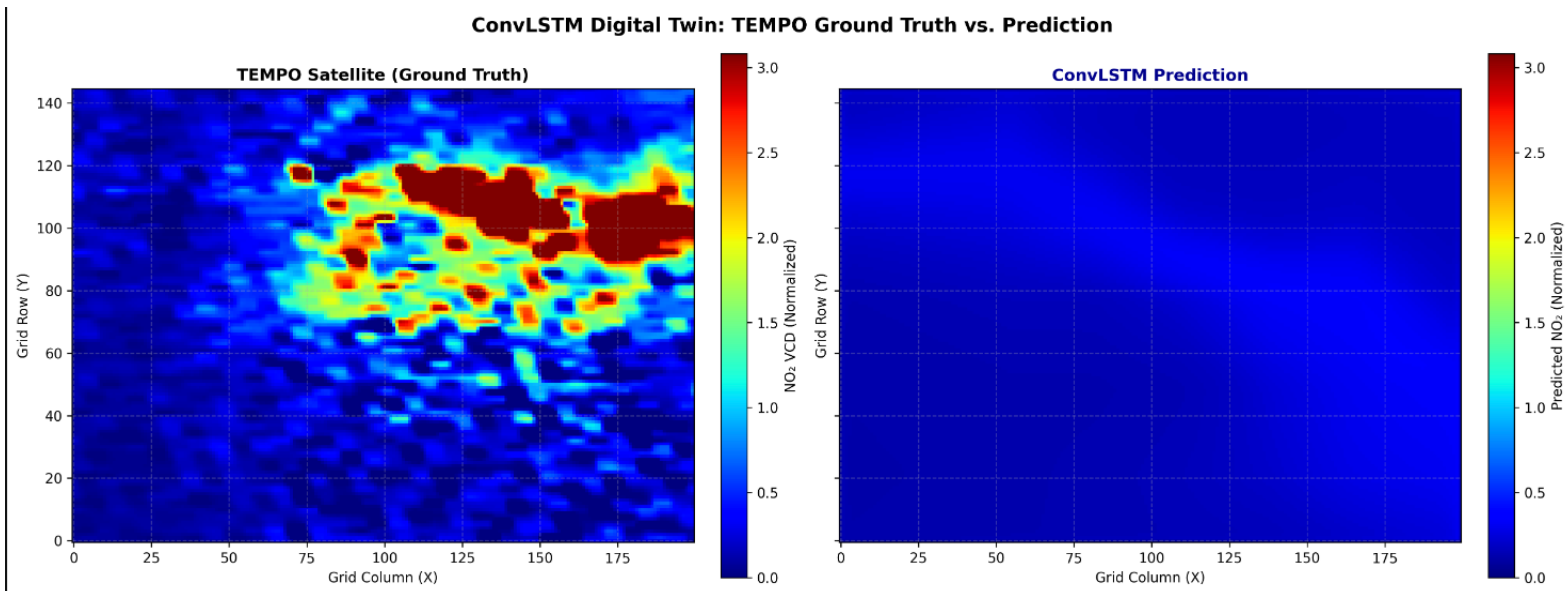
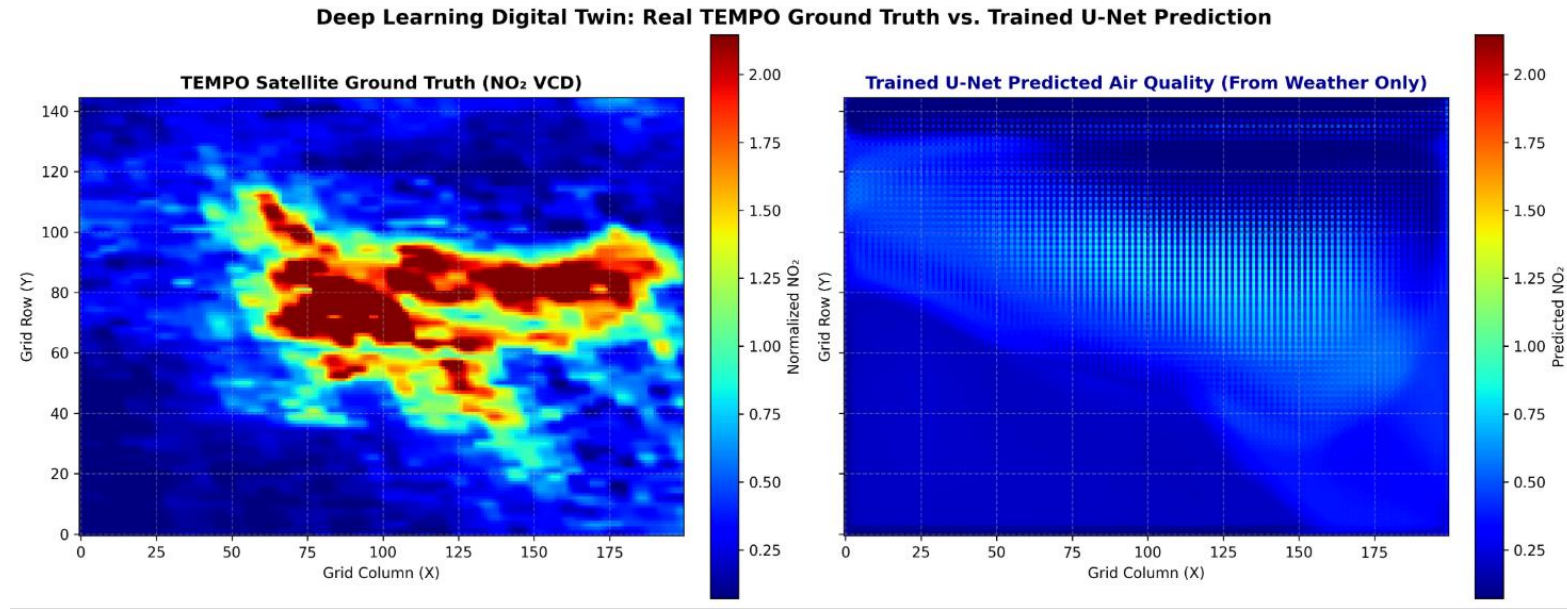
Land & emissions (6)

traffic_proximity_index
distance_to_major_road_km
pop
elev
distance_to_coast_km
wc_built

Time (4)

hour_sin
hour_cos
day_of_week_sin
is_weekend

Spatial Nowcasting attempts using other models: U-net and ConvLSTM



03 Methodology & Design

LA Basin

High Station Density,
Urban & Traffic
Sea Breeze

California

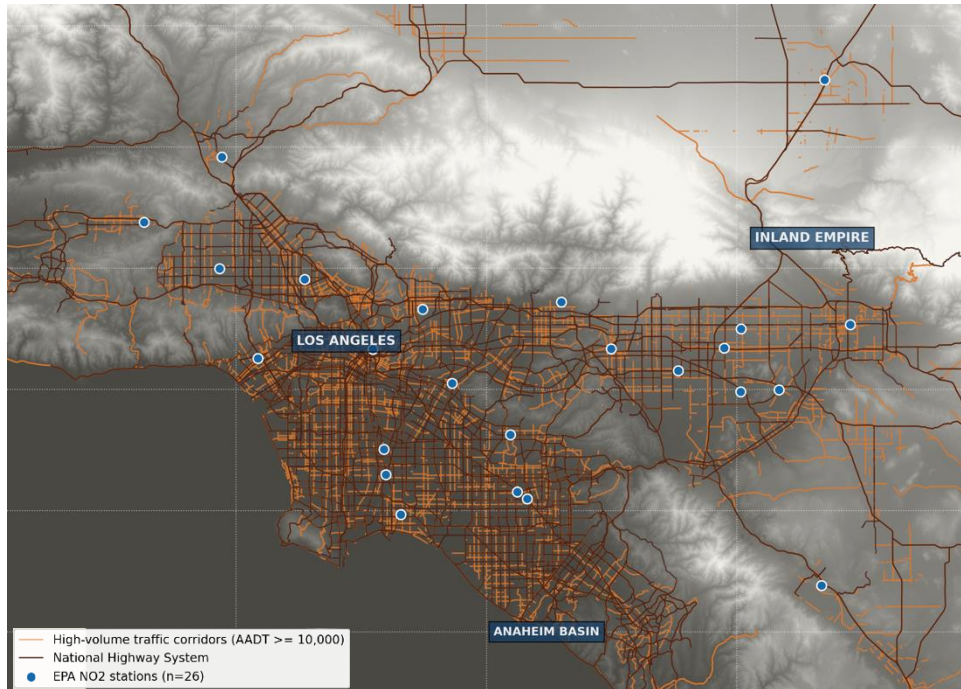
Complex Terrain
Mix of Urban, Valleys,
and Rural Areas

Western US

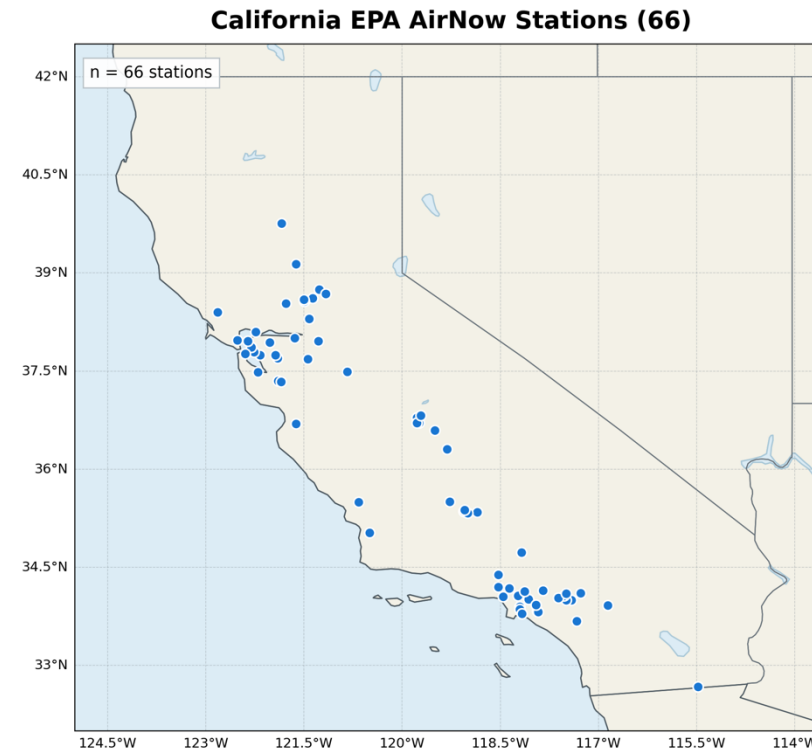
Lack of Ground Truth
Massive scale
Extreme Terrain

Can XGBoost be trained in a dense urban traffic area (LA), to complex valleys (CA) and vast sparse regions (Western US)?

Los Angeles

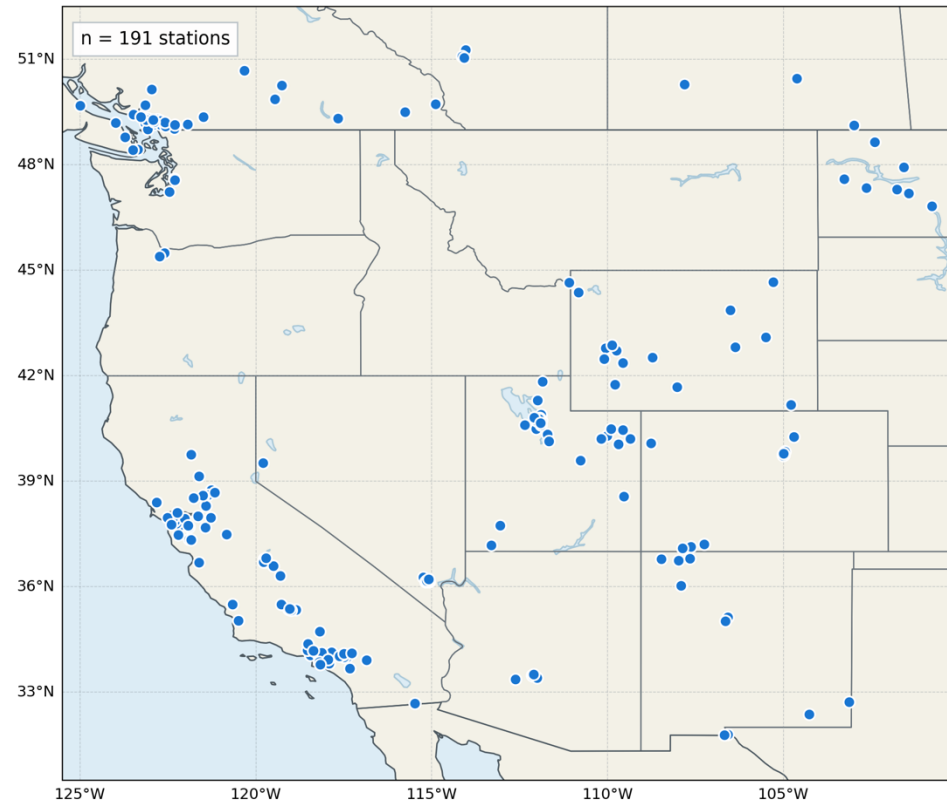


California



Western US

Western US AirNow Stations Used in Training



We Are Running Three Tests:

#1 Spatial Held Out Test

(Take out 20% of Stations from Training w/ K-means 5)

#2 Temporal Held Out Test

(Take out 3 months of 14 months from Training)

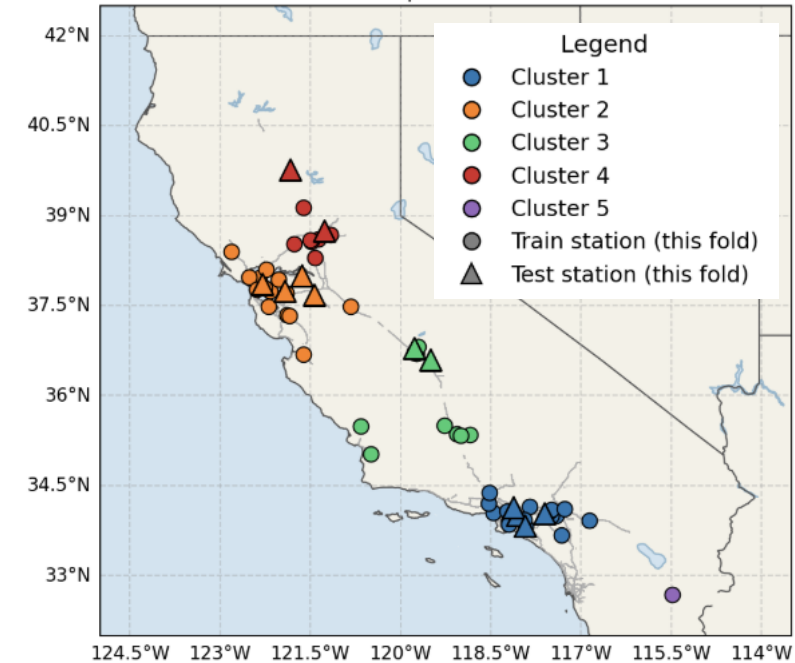
Train July 2023 - June 2024;

Test July 2024- Sept 2024)

#3 Temporal Held Out Test + Persistence

What if we add two predictors that ask what the NO_2 of the previous day and the last hour are?

#1



#2

TRAINING
July 2023 - June 2024
11 months

TESTING
July 2024 -
September 2024
3 months

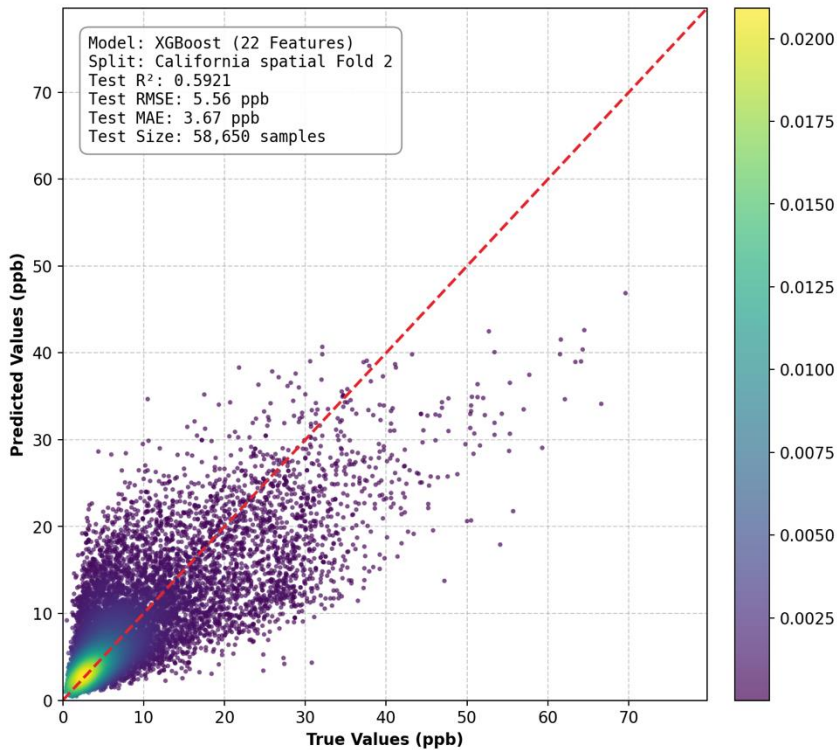
Training Testing

04 Results

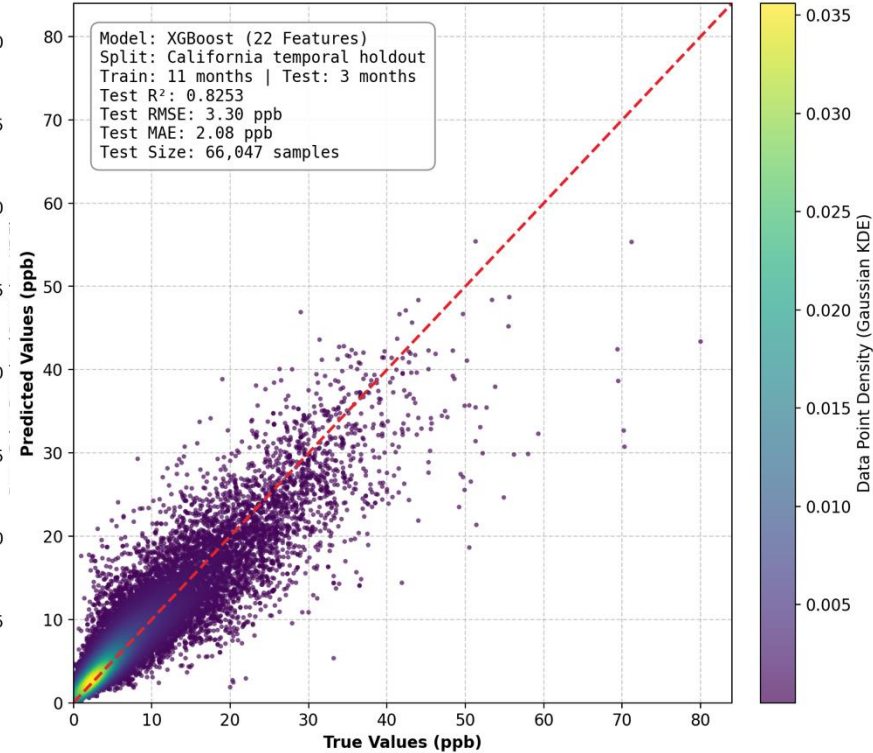
Region	Test	R^2	MAE (ppb)	RMSE (ppb)	Bias (ppb)
LA Basin	Spatial holdout	0.536	4.78	6.55	+0.09
LA Basin	Temporal holdout	0.792	2.94	4.31	-0.04
LA Basin	Temporal + persistence	0.909	1.80	2.84	+0.13
California	Spatial holdout	0.592	3.55	5.39	+0.54
California	Temporal holdout	0.825	2.18	3.63	-0.13
California	Temporal + persistence	0.901	1.39	2.49	+0.06
Western US	Spatial holdout	0.530	3.65	5.71	+0.20
Western US	Temporal holdout	0.773	2.07	3.37	-0.01
Western US	Temporal + persistence	0.889	1.32	2.35	+0.05

Visualization of R^2

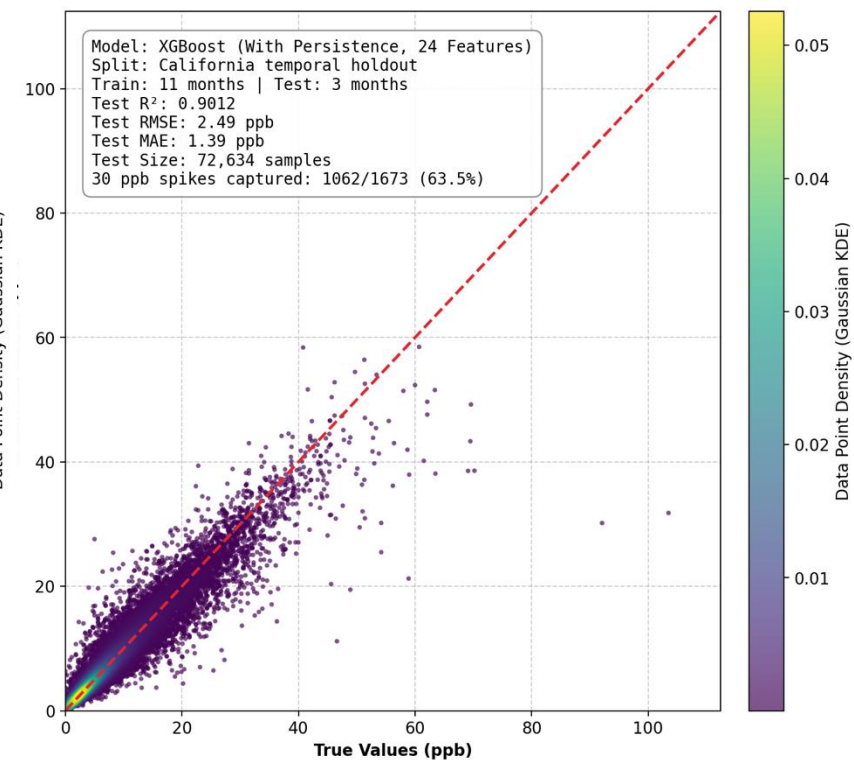
California Spatial Holdout XGBoost (Fold 2)
 $R^2 = 0.5921$



California Temporal Holdout XGBoost
 $R^2 = 0.8253$



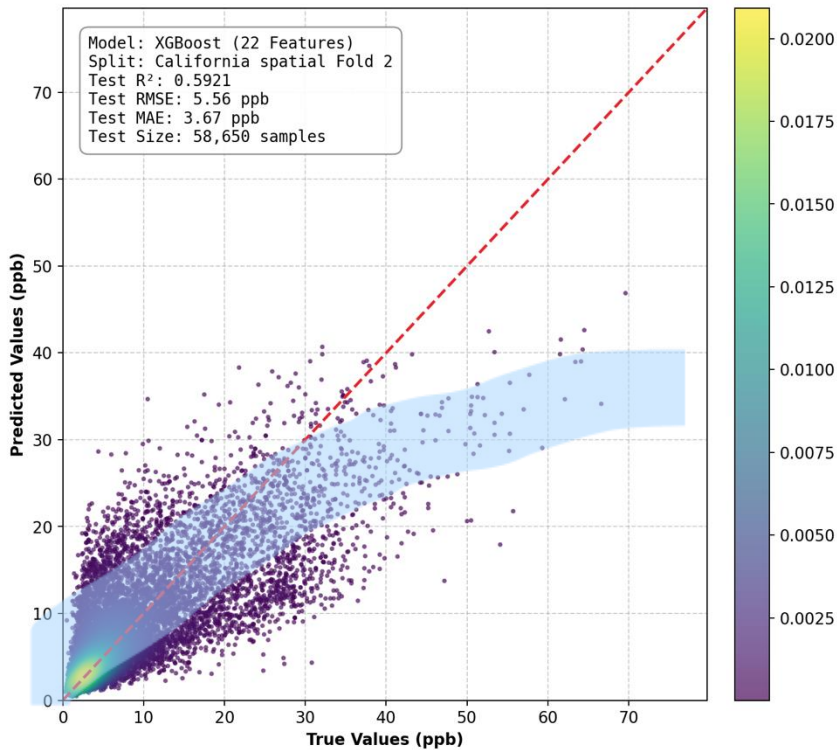
California Temporal Holdout XGBoost — With Persistence
 $R^2 = 0.9012$



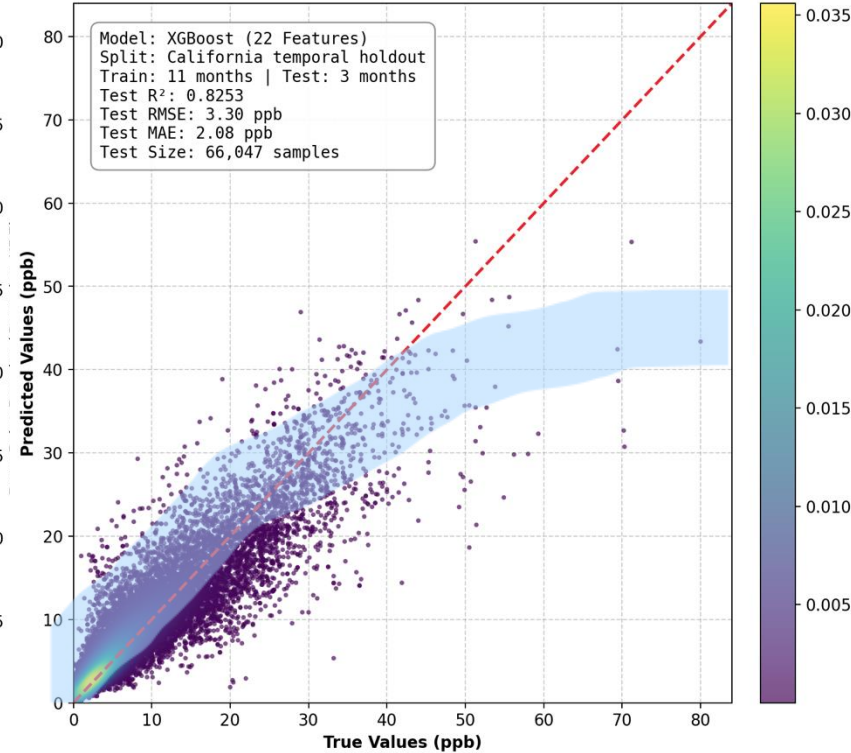
Machine learning does very well with *low-high values*, but expectedly, not well with extreme values

Visualization of R^2

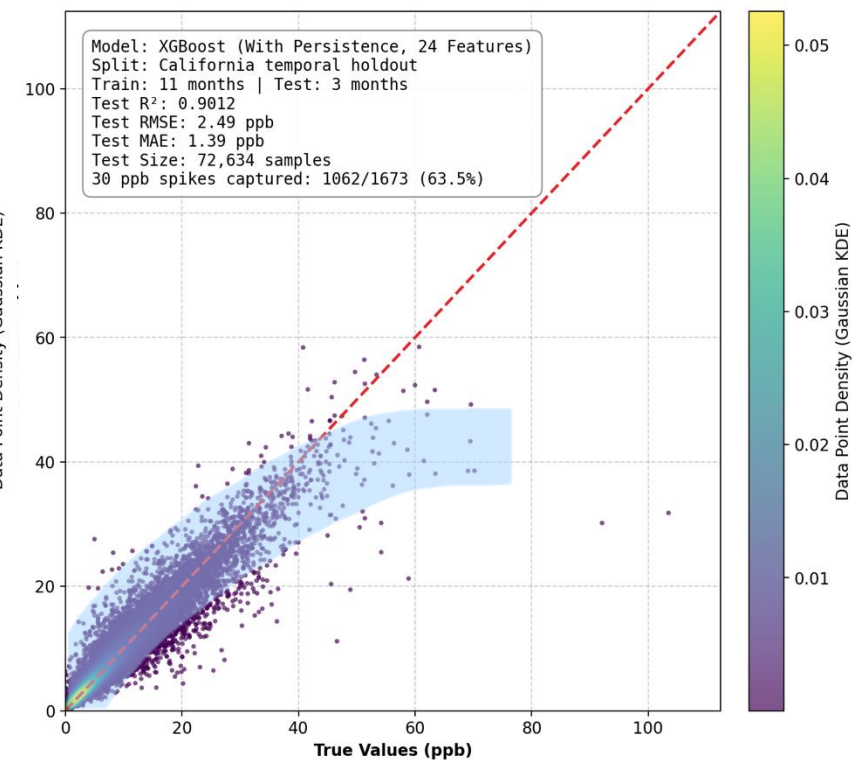
California Spatial Holdout XGBoost (Fold 2)
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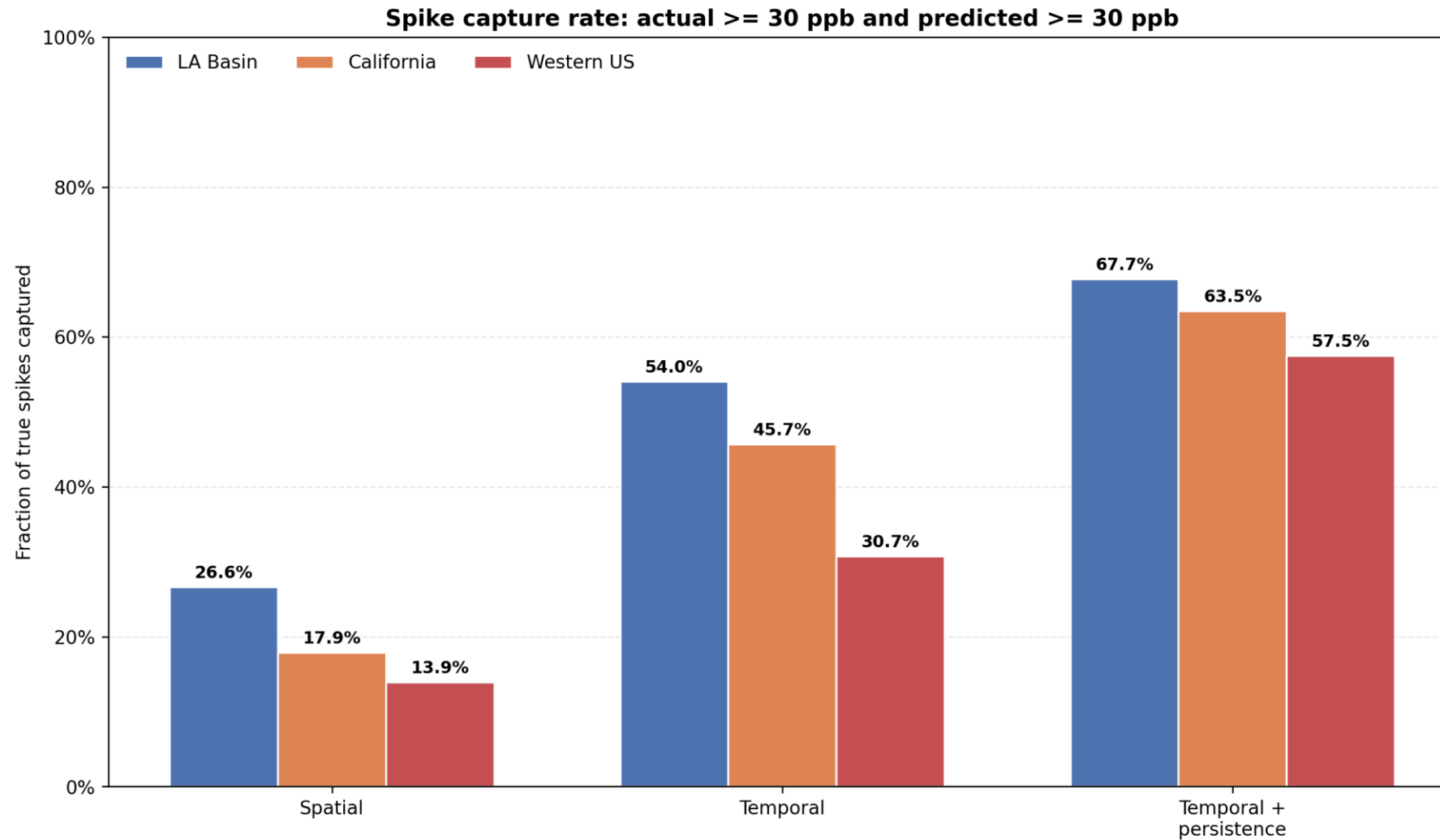
California Temporal Holdout XGBoost — With Persistence
 $R^2 = 0.9012$



Machine learning does very well with *low-high values*, but expectedly, not well with extreme values

Spatial Limits & Extreme Spike Diagnostics

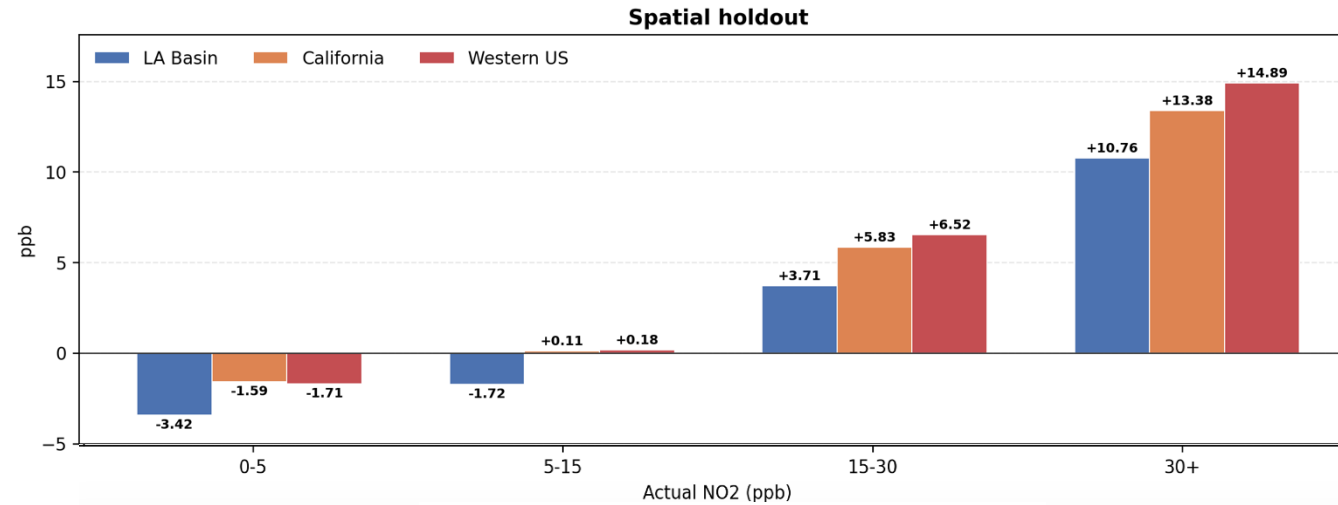
05 Analysis



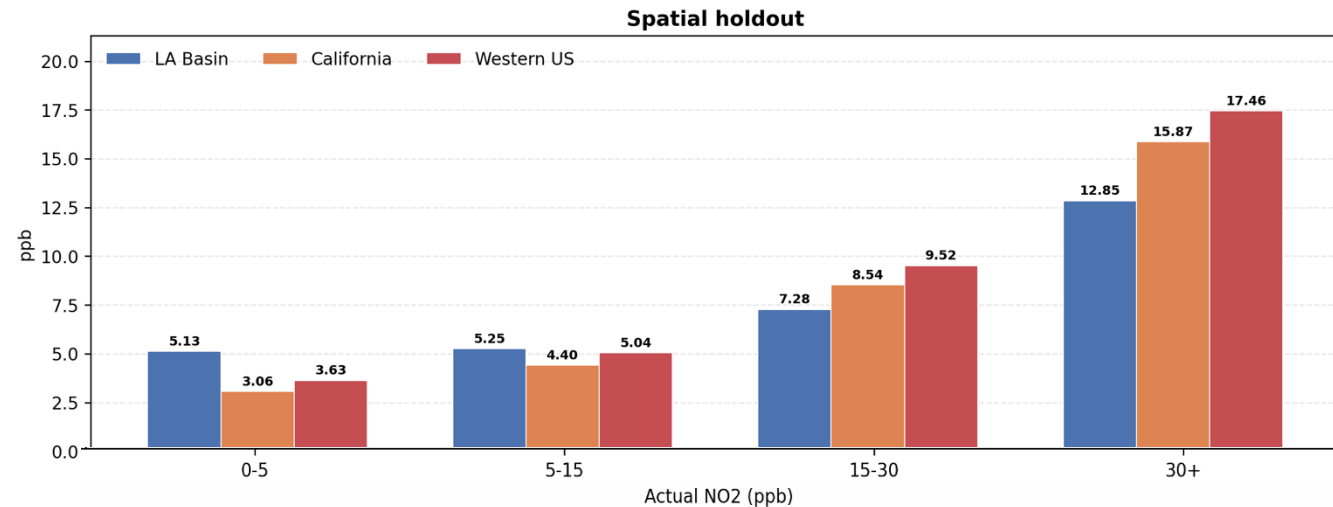
When is Spatial Good?

Bias by actual NO2 bin Positive values indicate underprediction

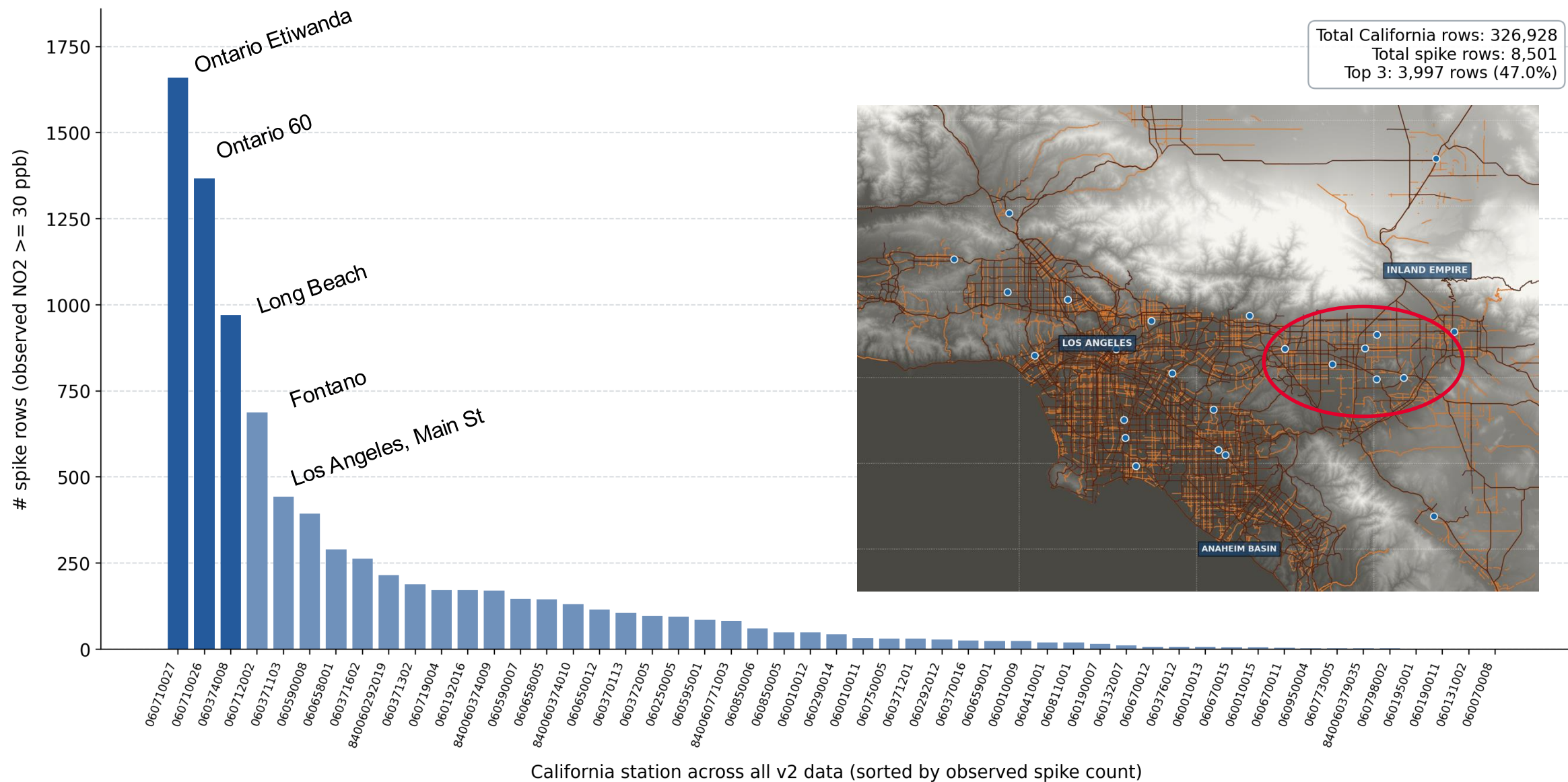
*Actual - Predicted



RMSE by actual NO2 bin *Model's errors, but gives extra weight to missed spikes

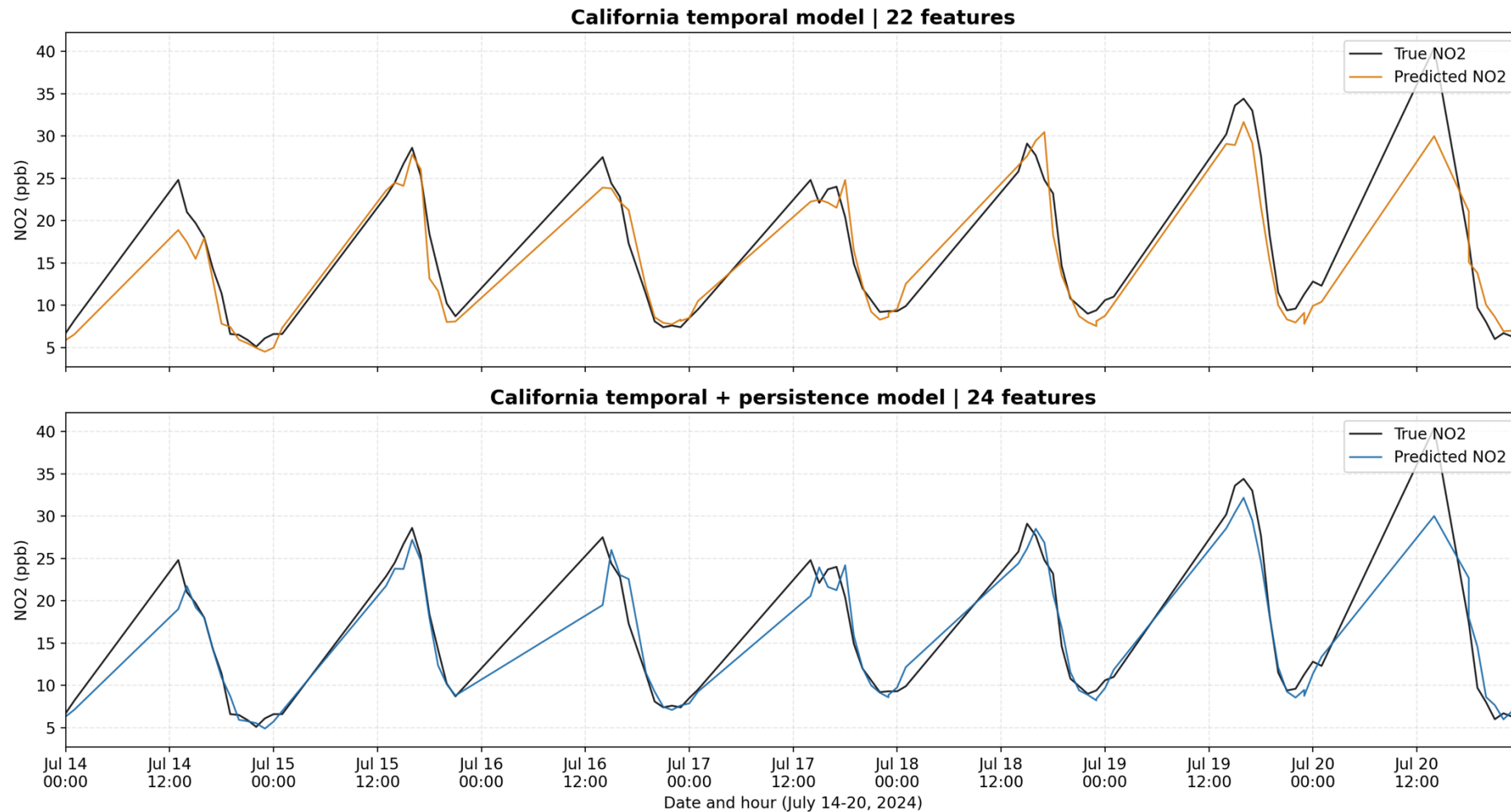


Are spikes concentrated? 51/66 stations spike; top 3 = 47% of all spikes



Fontano Location Time Series

Fontana EPA station: hourly California temporal predictions | July 14-20, 2024



06 Conclusion

Temporal & Persistence

#1 Short-term memory stabilizes predictions across all geographic scales (LA, CA, Western US).

I know WAY MORE about how algorithms “learn” and how to understand and question data!

I will be taking a machine learning class and statistics classes in the upcoming year!

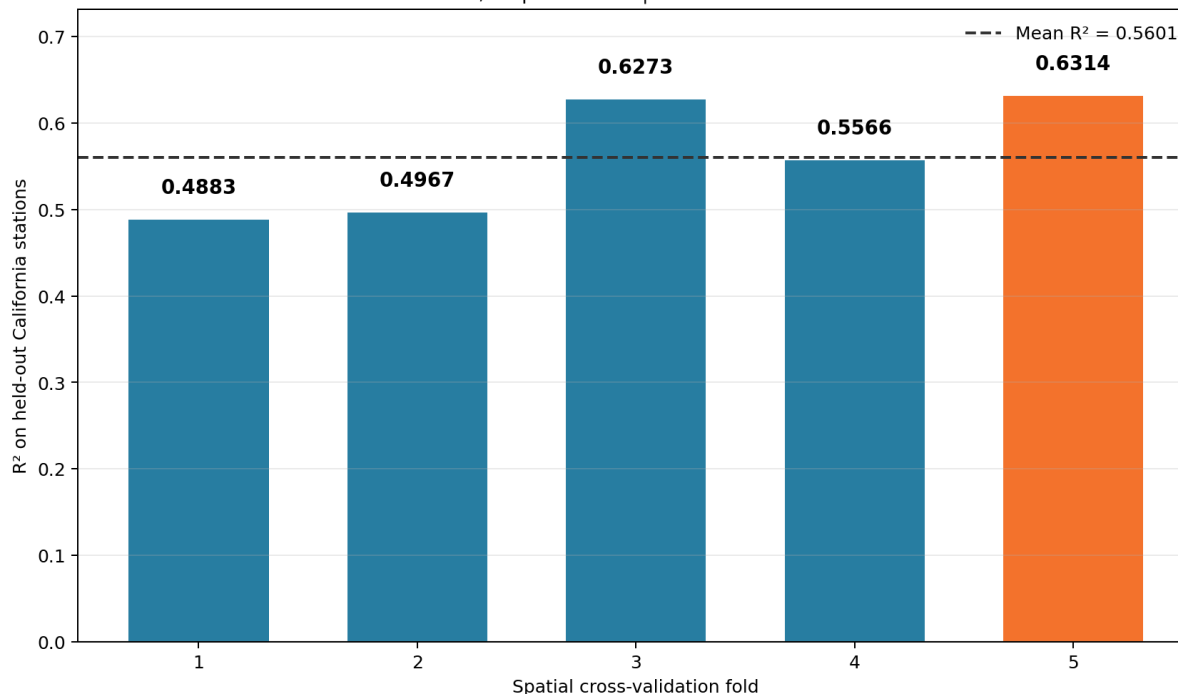
Spatial

#1 Boosting can't catch extreme localized pollution events.

Paper's best R^2 was 0.70, and we got 0.63!

California Spatial Holdout Performance

22 features, no persistence | 5 K-means-stratified folds



Future Plans: Scaling the NO2 Digital Twin

#1 Implement **Asymmetric Loss** to eliminate the **>30ppb underprediction bias**.



#2 Lean into **forecasting**, what's XGBoost's **capability to forecast**, can I add **Wind Highways** and **Mountain Ranges**?



#3 LEARN MORE! Instead of **XGBoost**, explore **Mamba** and **Graph Neural Networks (GNNs)** for trials?.

Questions?

Acknowledgments

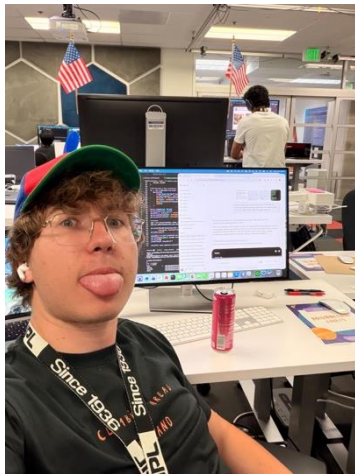
Kyo Lee
Michael Turmon
Tiago Stegun
Mircea Badescue
Luis Marchen
Olga Kalashnikova
Everyone I've had a coffee chat with
JPL Friends!

Vasya Easley | Section 341G
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*Image credit: NASA,
Vecteezy*





Thank You!

Feel free to ask more afterwards!



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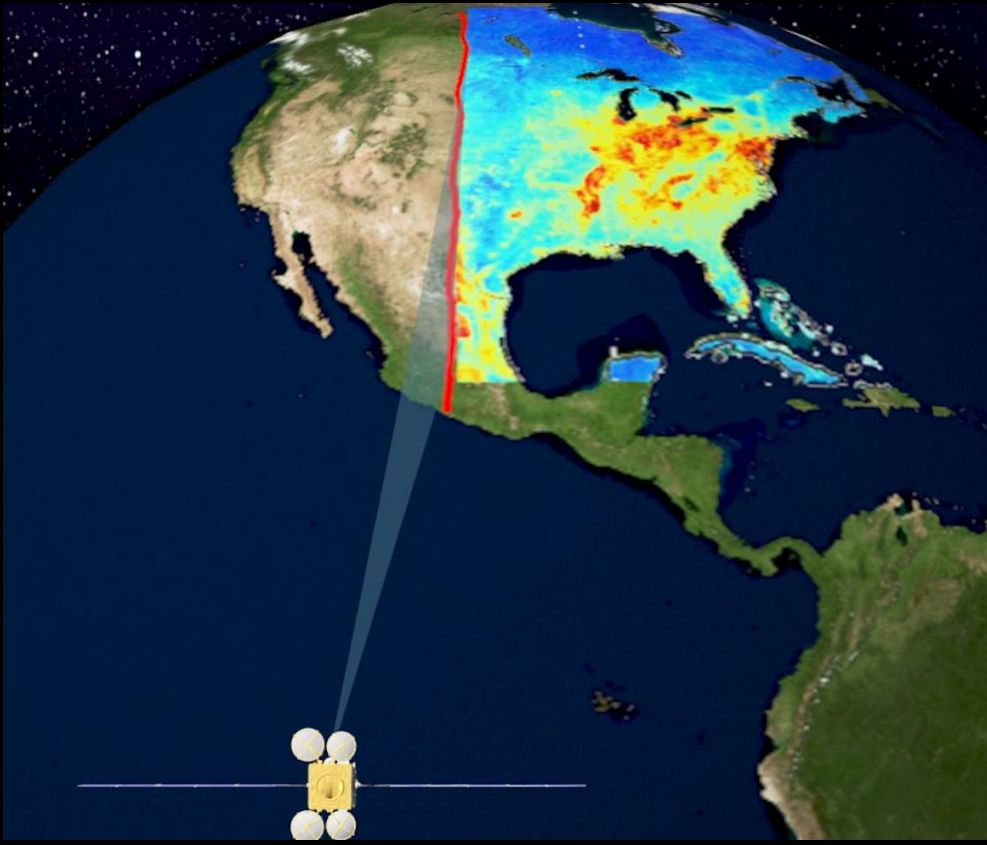


Image credit: NASA EARTHDATA

07

Recap

08

Reasons and Motivations

09

Methods

10

Results

11

Future Direction

12

Questions

Recap

- Prior project: Air-Now NO₂ Data → Air-Now NO₂ (no2-forecasting)
- Used only ground-station information
- Performance was limited
- Results were not strong enough

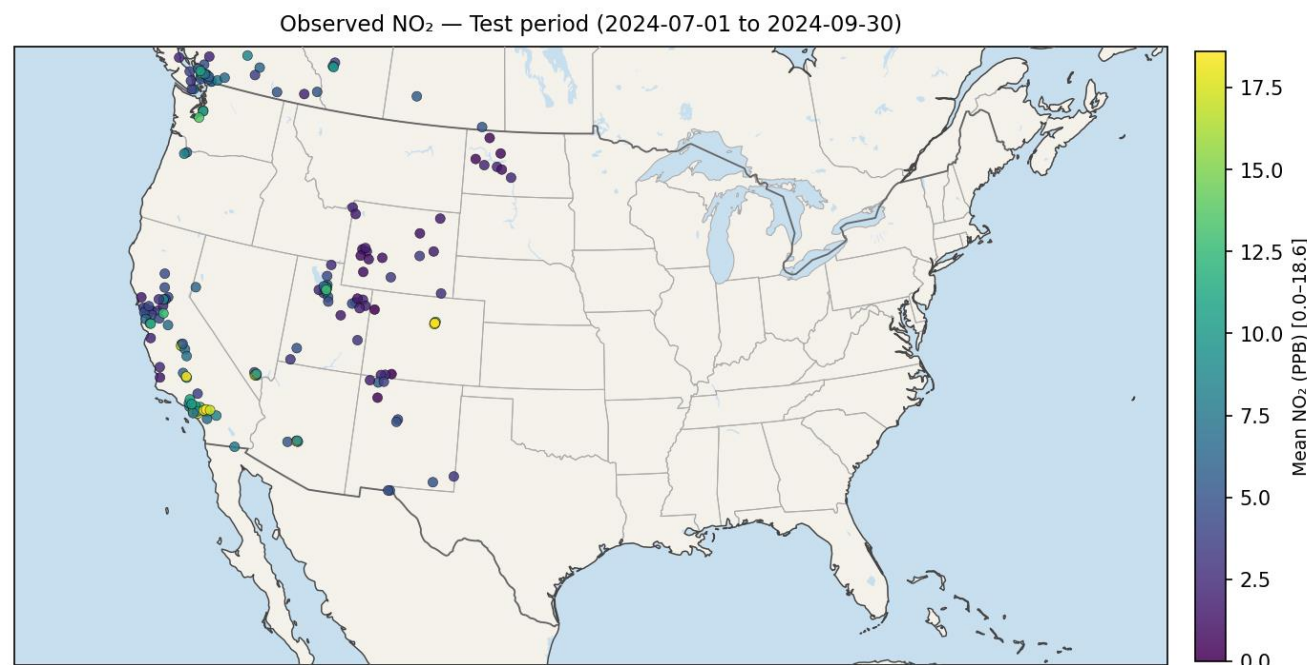
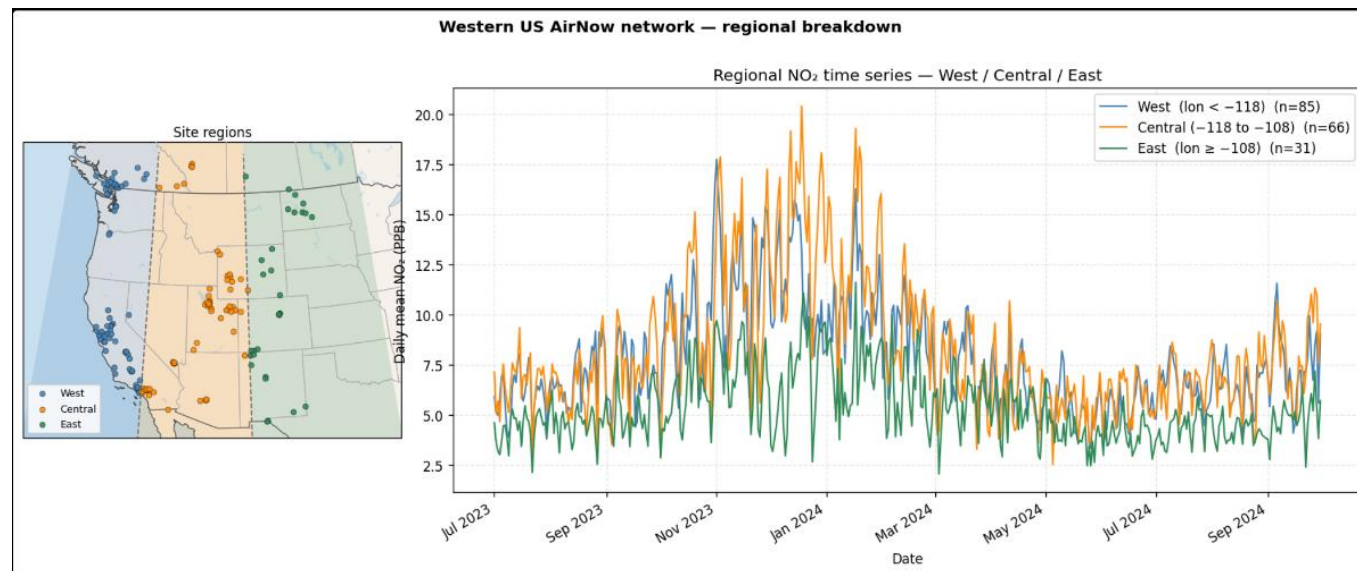


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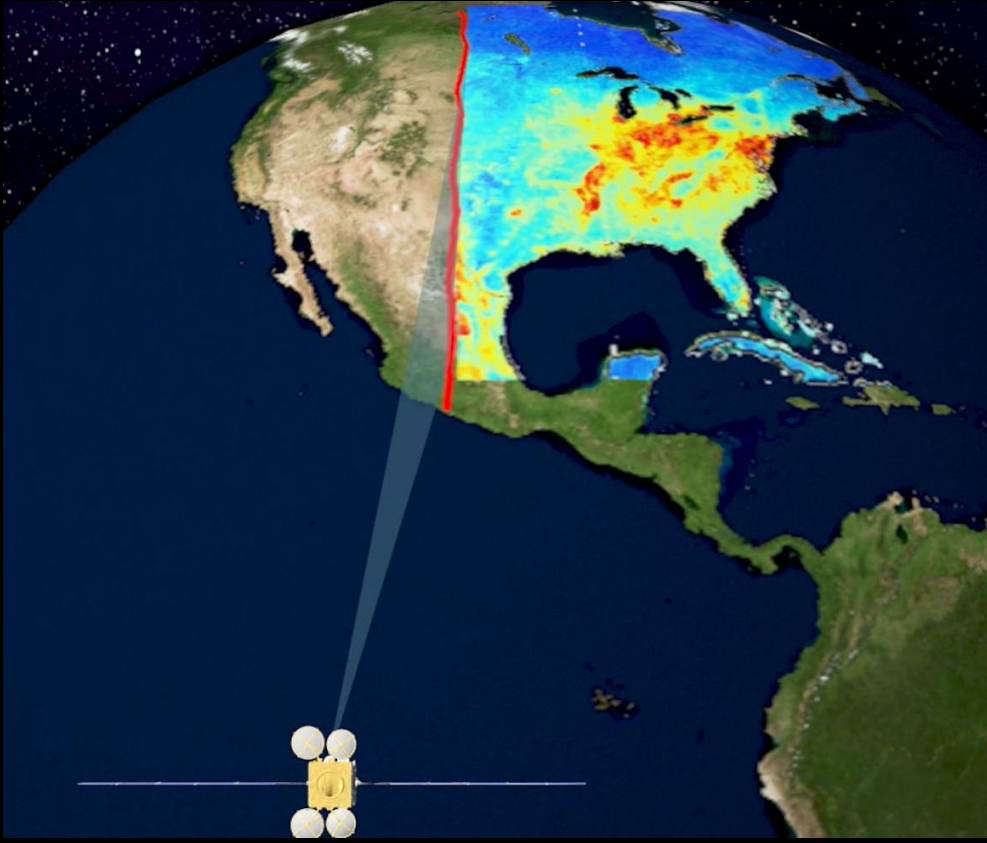


Image credit: NASA EARTHDATA

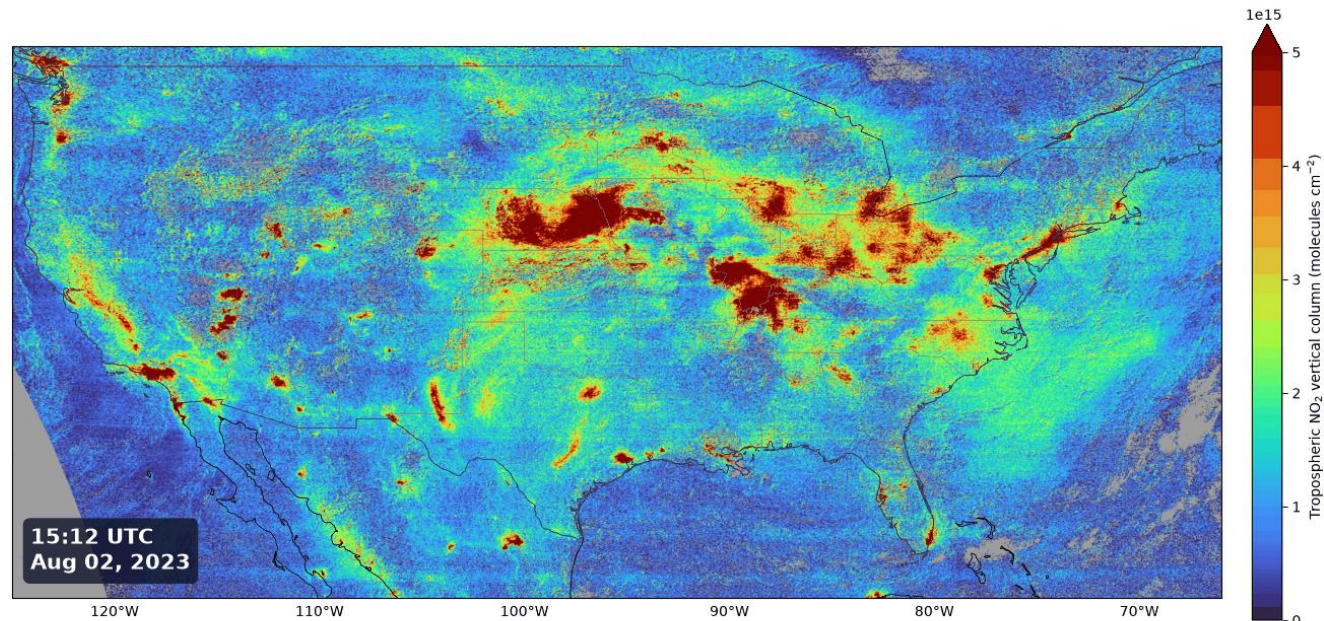
- 07** Recap
- 08** Reasons and Motivations
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Reasons and Motivations

- Air-Now NO₂ data was not enough
- Added TEMPO NO₂ Data as a predictor
- GNN Transformer & Mamba took too long to run

Objectives

- Add TEMPO satellite information
- Add nearby-station spatial context
- Keep model training practical while having spatial + temporal features
- Forecast 1 hr ahead for NO₂ levels



TEMPO NO₂_L3_V04 (Aug 2nd)

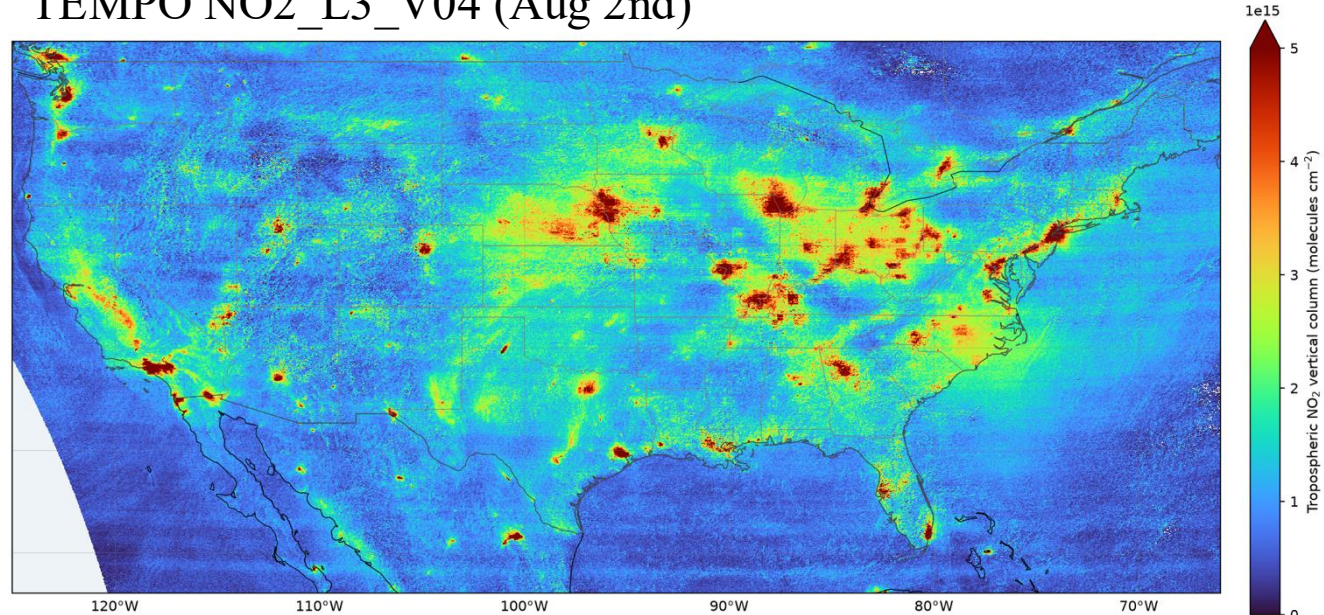


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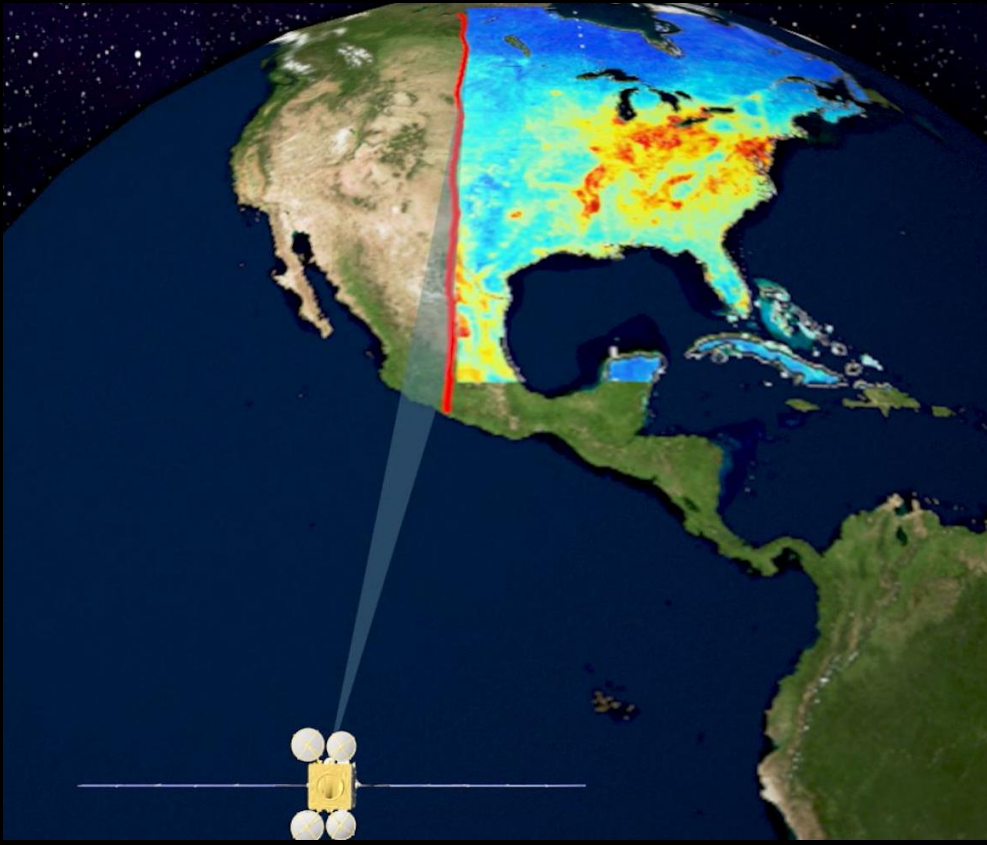


Image credit: NASA EARTHDATA

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Training Setup

- Train: 2023-07-01 to 2024-06-01
- Validation: 2024-06-01 to 2024-07-01
- Test: 2024-07-01 to 2024-10-01



27 predictors

TEMPO -> AirNow NO2 — full feature set, grouped by category

TEMPO satellite

n = 6

tempo_no2
tempo_no2_lag_1
tempo_no2_lag_3
tempo_no2_roll_mean_3
tempo_no2_roll_std_3
tempo_no2_delta_prev

Spatial

n = 2

latitude
longitude

Calendar & time

n = 7

hour_of_day
day_of_week
is_weekend
month
season
doy_sin
doy_cos

Meteorology (ERA5)

n = 10

sp
u_1000hpa
v_1000hpa
t_1000hpa
d2m
tcc
u10
v10
t2m
blh

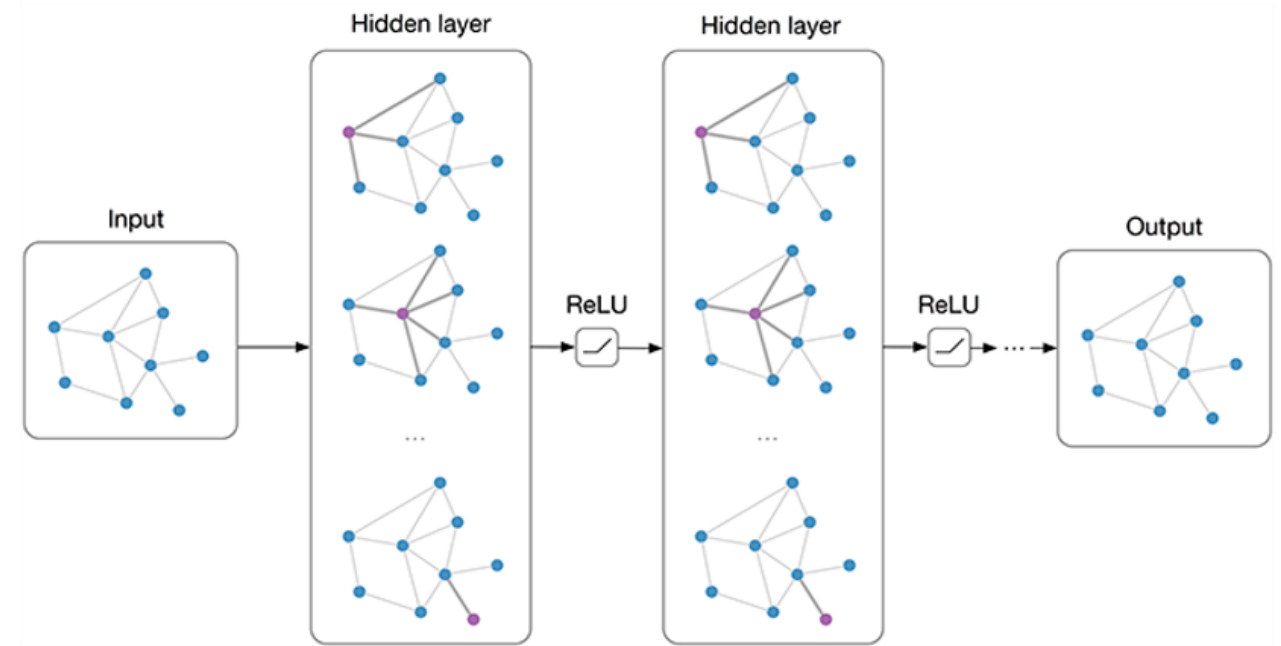
Traffic

n = 2

road_density
traffic_aadt

What is a GNN?

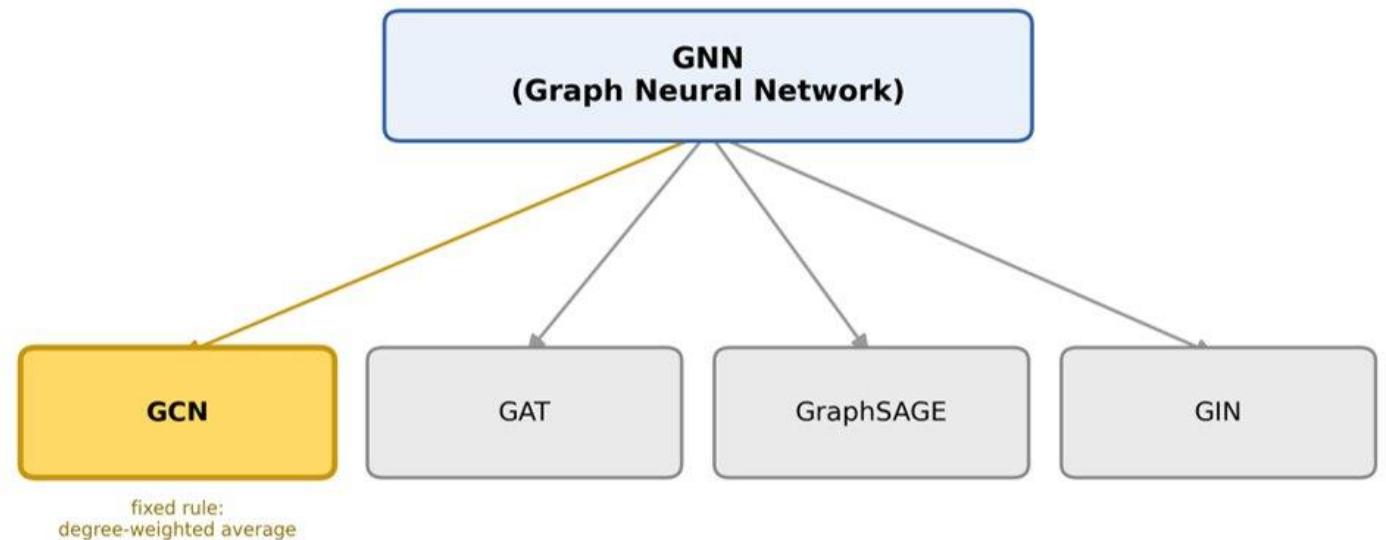
- Graph Neural Network
- Works on graph data
- Nodes learn from neighbors
- Repeats over layers



<https://tkipf.github.io/graph-convolutional-networks/>

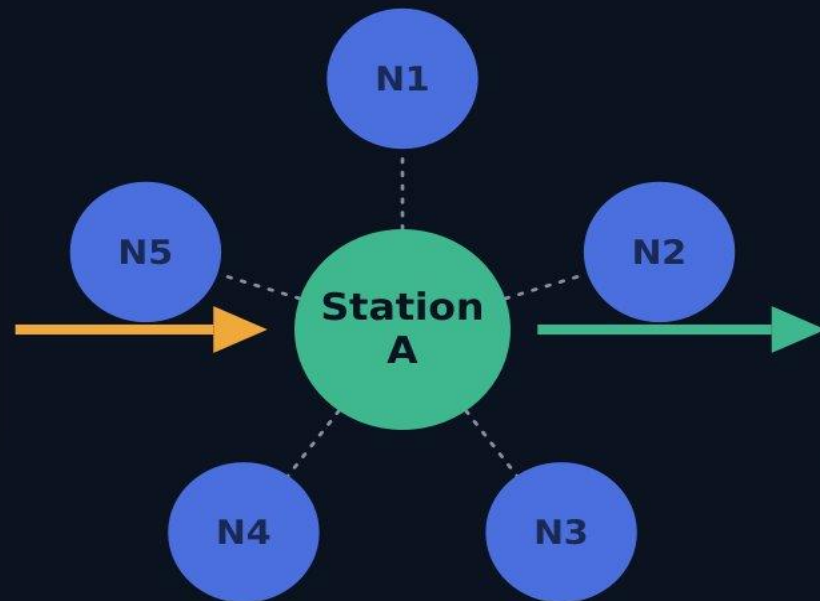
What is a GCN?

- GCN – Graph Convolutional Network
- A specific type of GNN
- Works on data shaped like a "network"
 - nodes connect to other nodes
- Each node updates itself by averaging



INPUT - per node, per hour

Station's own parameters



OUTPUT

One number: predicted ground-level NO₂ (ppb) at Station A, this hour

GCN Transformer & GCN Mamba

- Transformer: an attention-based model --> looks at all 24 hours at once
- Mamba: a state-space based model --> reads the 24 hours in order, one at a time

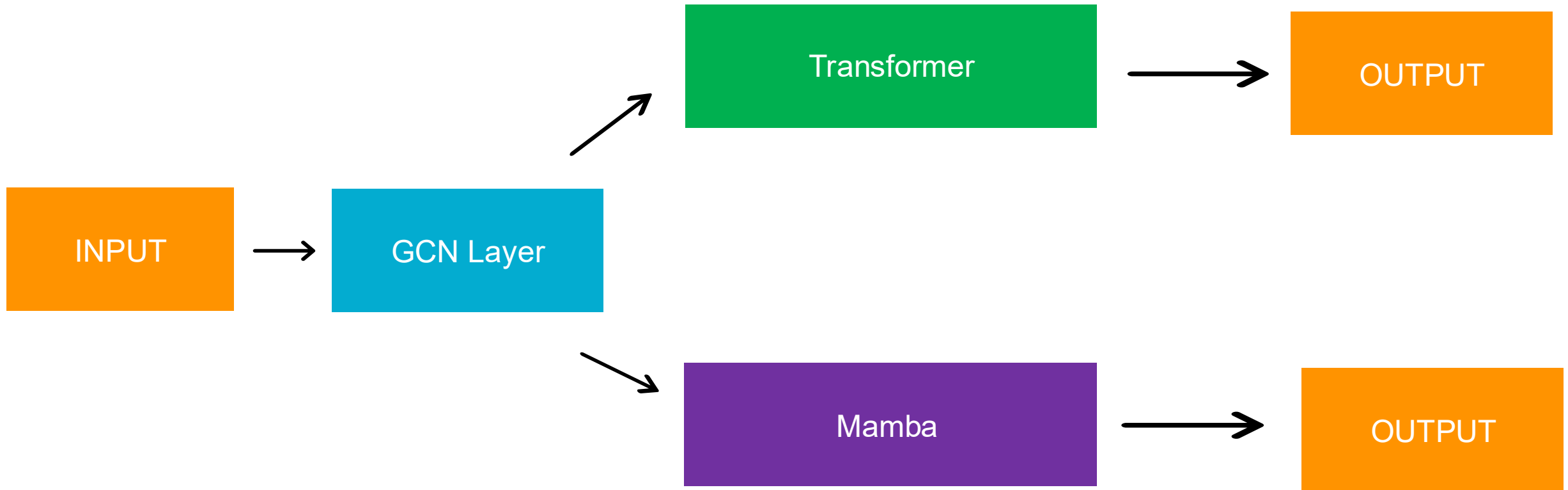


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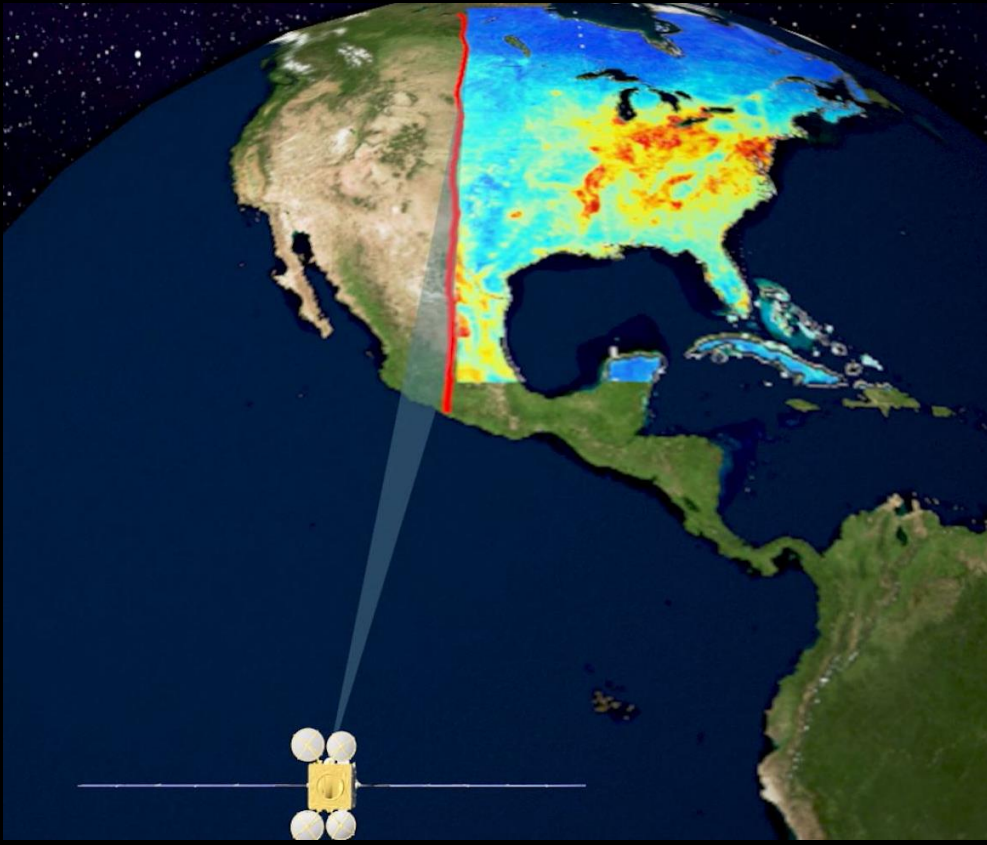
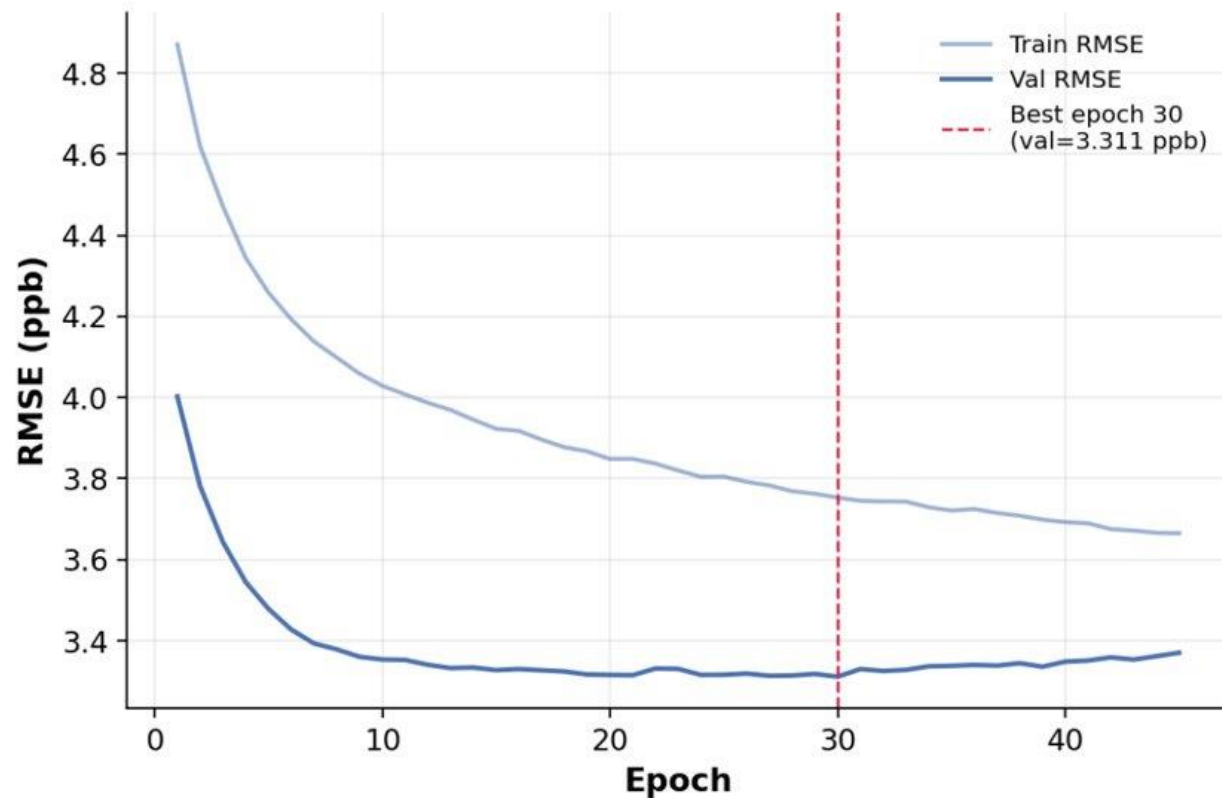


Image credit: NASA EARTHDATA

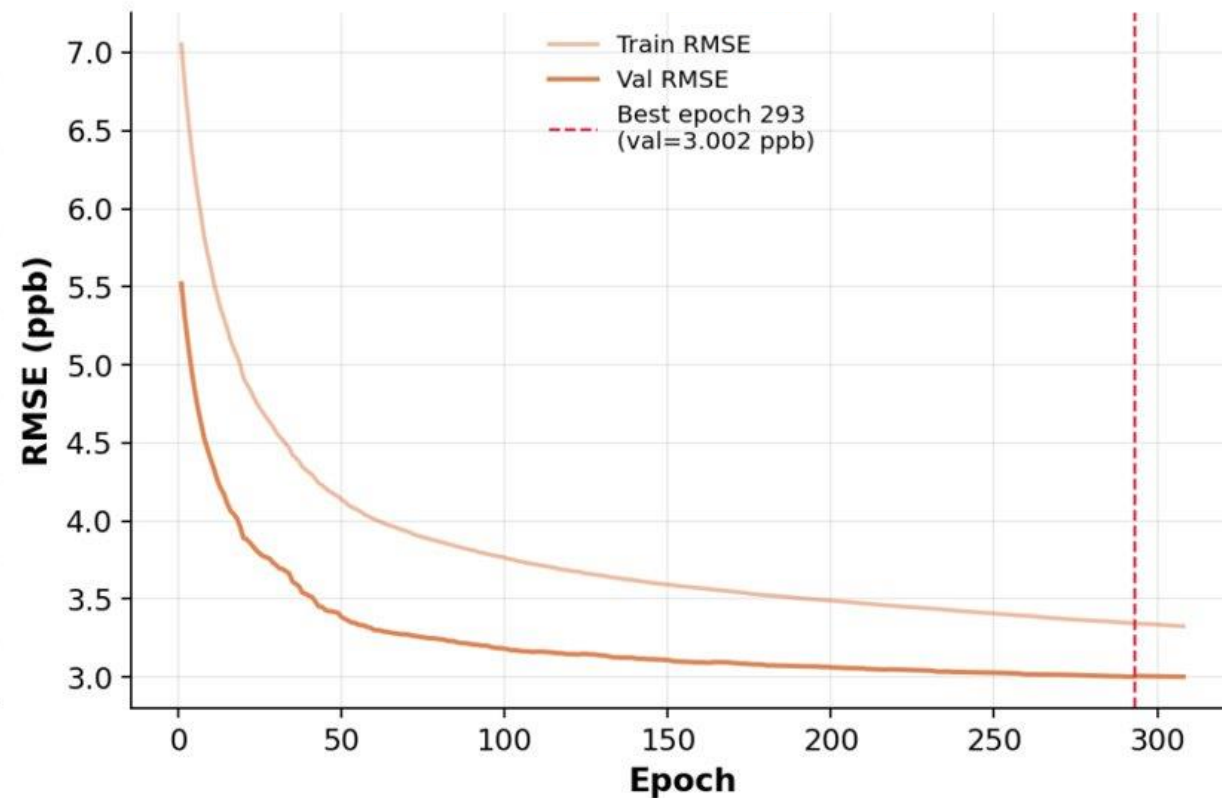
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Training curves: epoch vs. RMSE

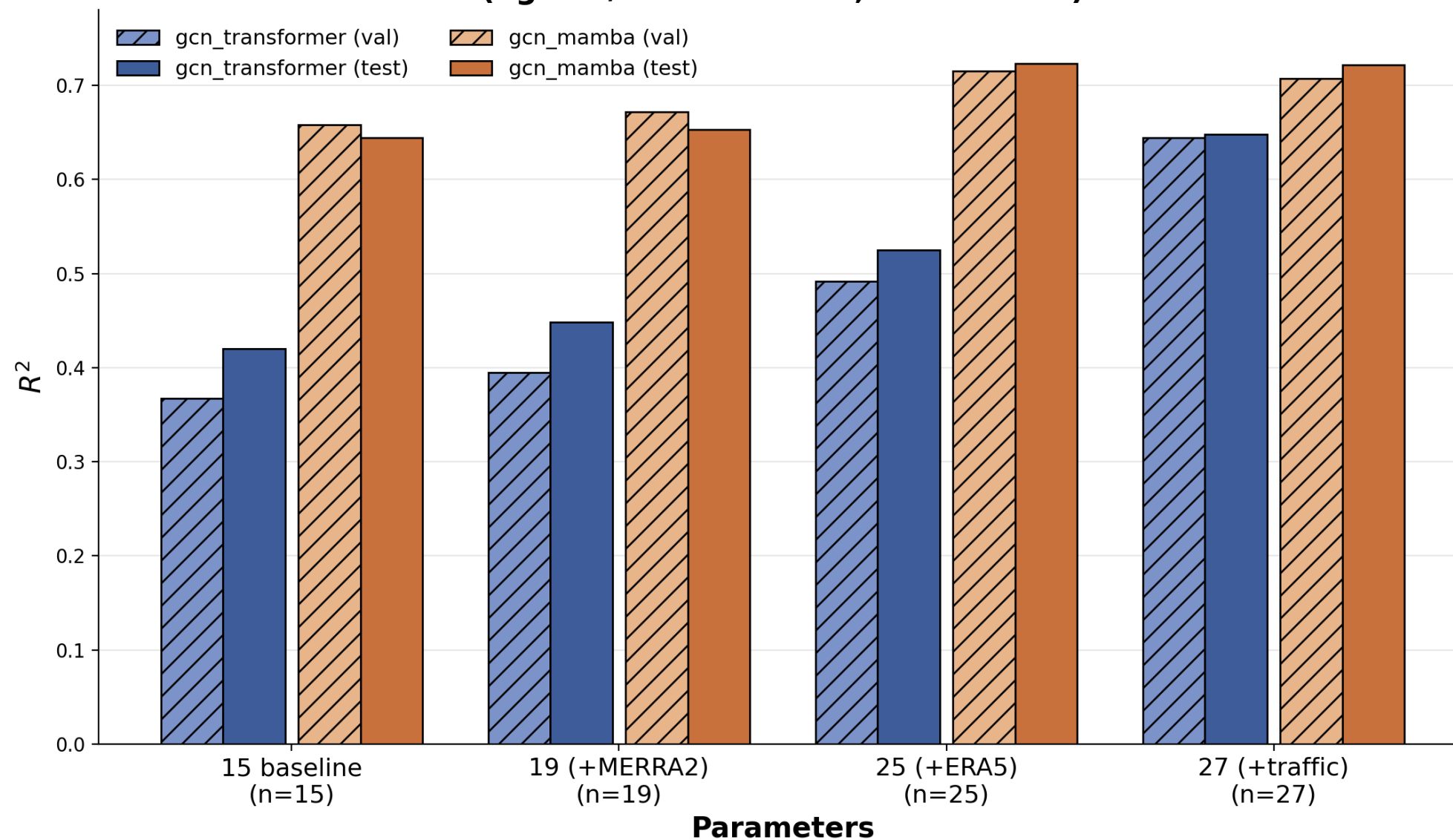
GCN-Transformer



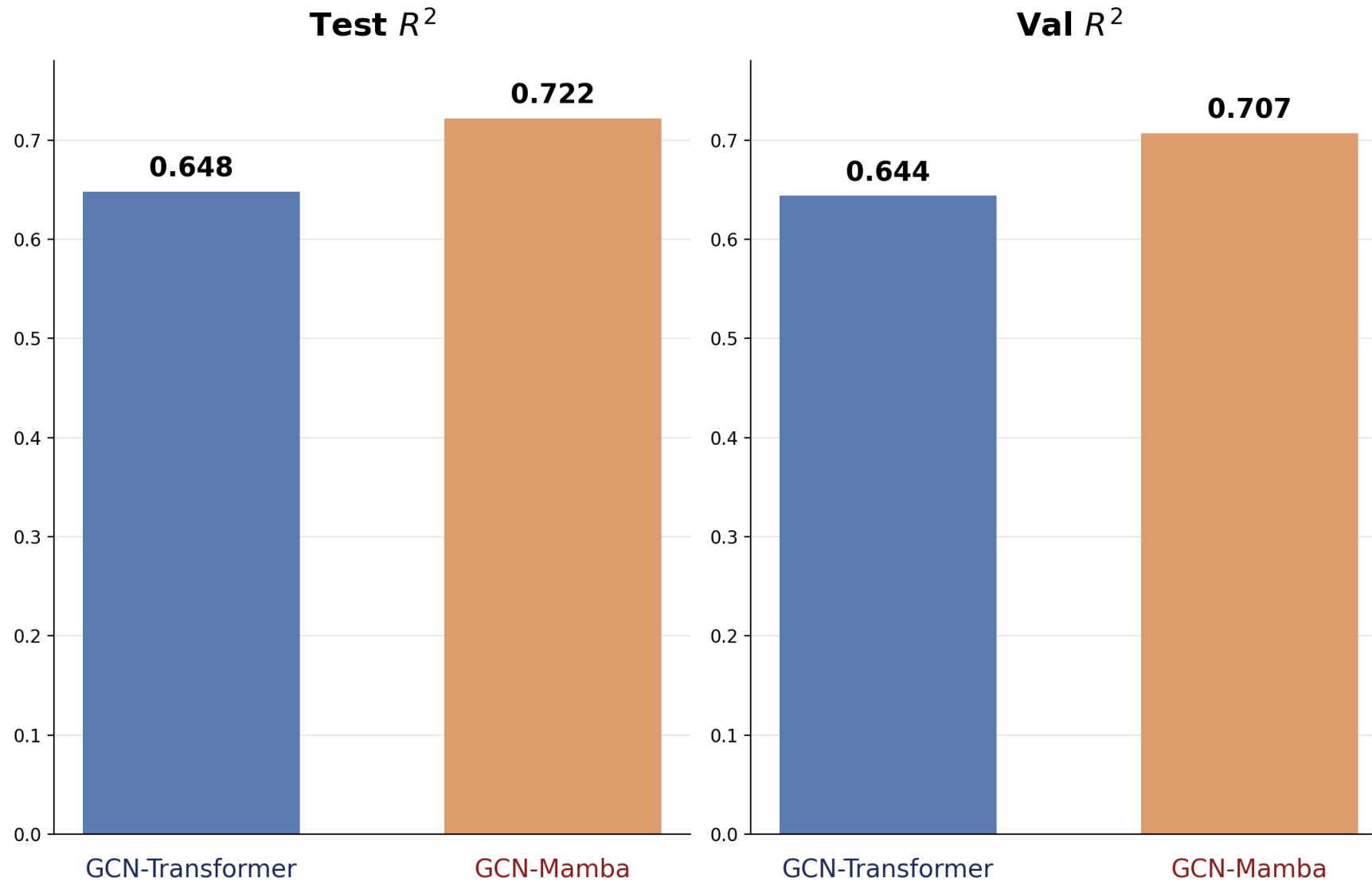
GCN-Mamba



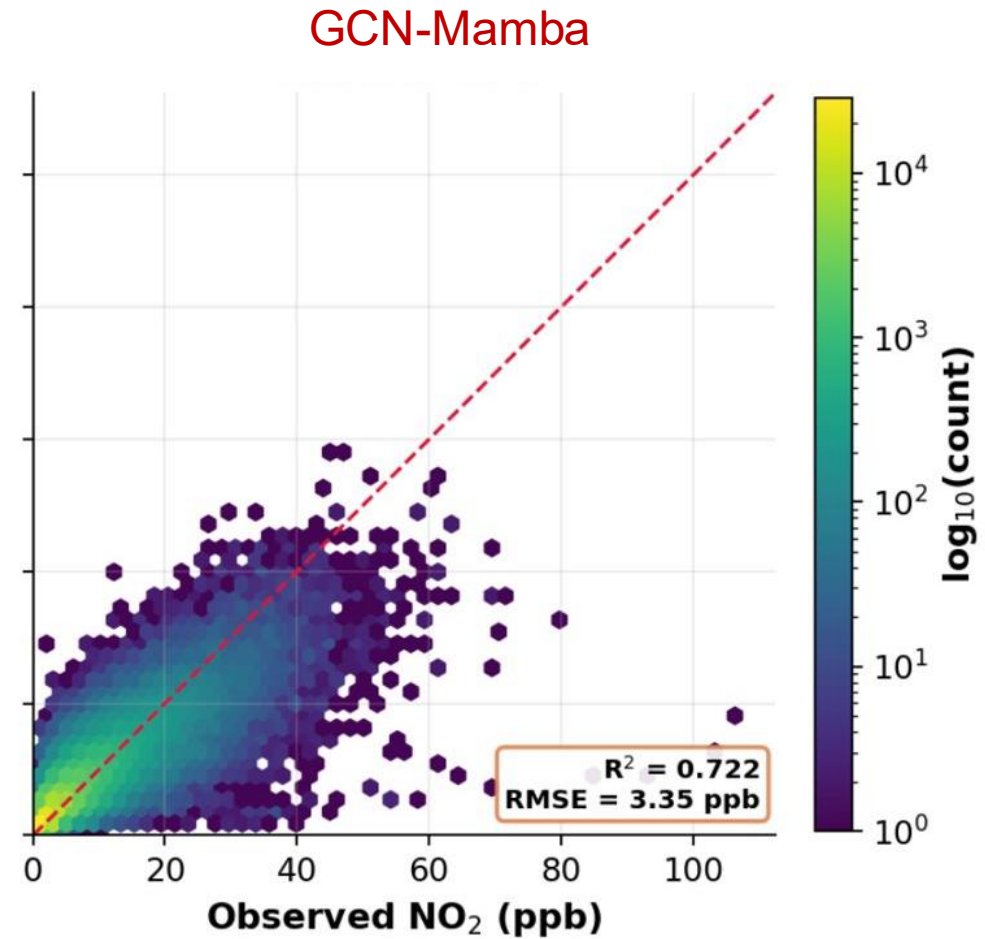
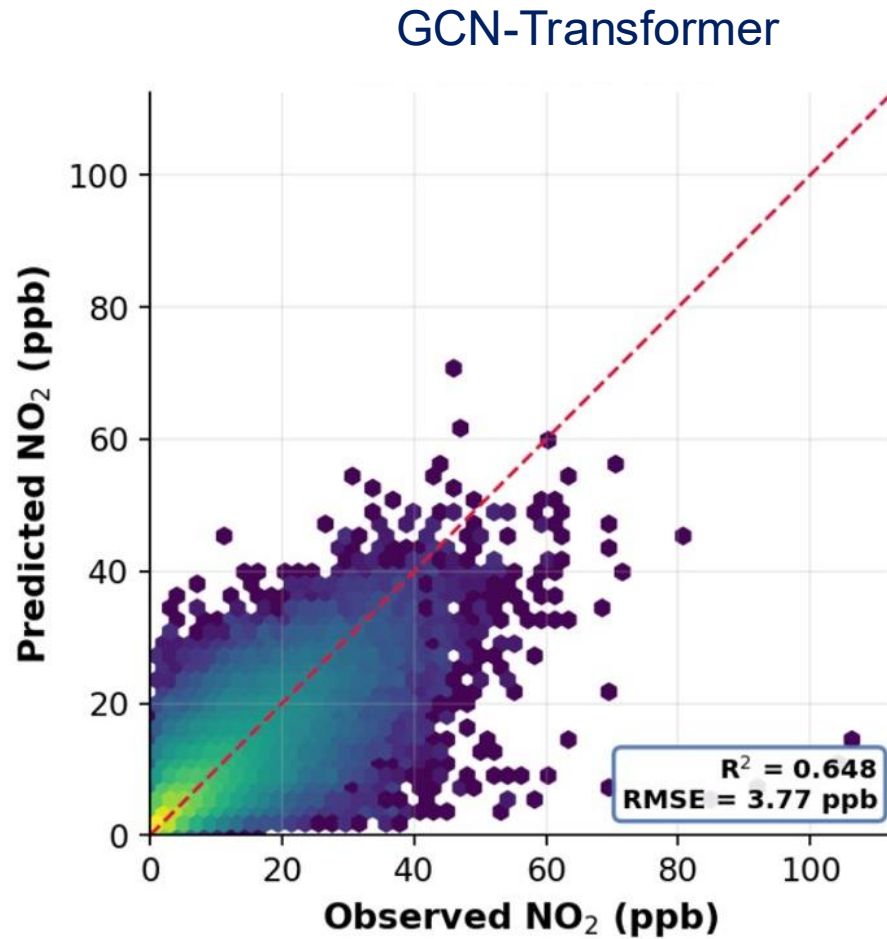
Feature ablation: R^2 by tier and model (lighter/hatched = val, solid = test)



GCN Mamba and GCN Transformer Final Metrics



GCN Transformer vs. GCN Mamba: Predicted Vs. Observed NO₂



Vasya vs Isabelle (Western US)

Vasya

	Model	R ²	RMSE (ppb)
●	Spatial	0.530	5.71
●	Temporal	0.773	3.37
●	Temporal + Persistence	0.889	2.35

- *Only usable at AirNow Stations
- *Usable Everywhere

Isabelle

	Model	R ²	RMSE (ppb)
●	GCN-Transformer	0.648	3.77
●	GCN-Mamba	0.722	3.35

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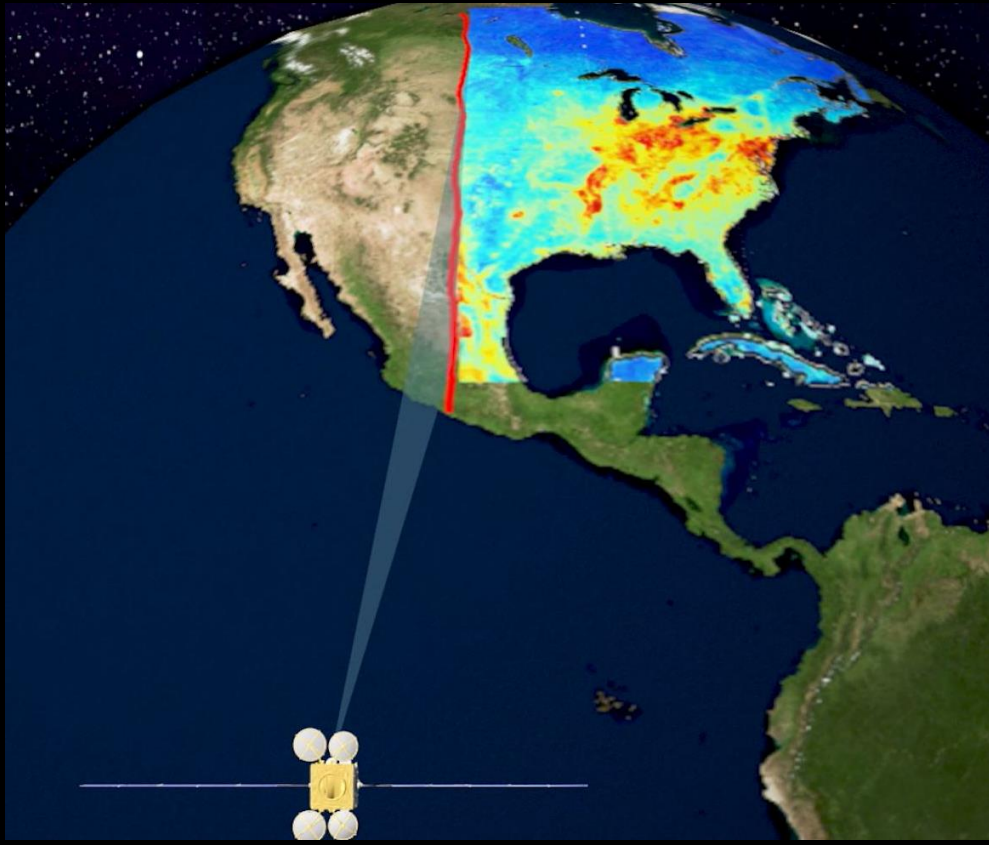


Image credit: NASA EARTHDATA

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Future Direction

#1 Have a bigger dataset

- there was a lot of missing data
- For better seasonal coverage and more spikes

#2 Use more complex models/try other variations of the GNNs

- Maybe the GCN was too simple for better results

Questions?



THANK YOU!

